

Fusion draft: SST-15/24/53 Thermodynamics and quantization

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About this fusion draft

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Chapter 1

SST-15 — Circulation-loop thermodynamics

Thermodynamics of a Circulation-Loop Gas in an Incompressible Fluid and the Classical Ultraviolet Problem

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Abstract

We investigate a classical analogue model for thermal radiation based on an incompressible, inviscid fluid populated by thin circulation loops (vortex rings) of finite core radius. Treating closed circulation loops as localized, finite-energy structures with effective mass M_{eff} , we construct a kinetic theory of a dilute loop gas in the canonical ensemble. The resulting partition function coincides with that of a classical ideal gas, and the equation of state $PV = Nk_{\text{B}}T$ follows in the standard way from the Helmholtz free energy. For an isotropic ensemble of ultra-relativistic or effectively massless excitations built from these loops, the relation $P = \frac{1}{3}u$ between pressure and energy density is recovered from the momentum flux tensor, relying only on three-dimensional isotropy and the relativistic relation $E = pc$. A finite core radius a and an upper bound v_{max} on the tangential speed inside the cores imply a kinematic maximum internal frequency $\omega_{\text{max}} = v_{\text{max}}/a$. If electromagnetic normal modes are identified with mechanically realizable excitations of this medium, only modes with $\omega \leq \omega_{\text{max}}$ are thermally accessible. The classical Rayleigh–Jeans integral is then automatically regulated, yielding a finite energy density $u \propto k_{\text{B}}T \omega_{\text{max}}^3$ without any modification of Maxwell’s equations. The resulting cutoff Rayleigh–Jeans spectrum does not reproduce the T^4 scaling of the Planck law and therefore does not replace quantum statistics. Instead, it provides an explicit mechanical example in which microstructure and finite rotational frequencies eliminate the classical ultraviolet catastrophe at the level of microstate counting, while leaving intact the standard macroscopic equations of motion and equations of state.

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1.1 Introduction

Classical thermodynamics and statistical mechanics provide a remarkably successful macroscopic description of fluids and radiation. The equation of state of an ideal gas,

$$PV = Nk_B T, \quad (1.1)$$

and the relation between pressure and energy density for an isotropic radiation field,

$$P = \frac{1}{3}u(T), \quad (1.2)$$

are textbook results derived from kinetic arguments and from the stress–energy tensor of relativistic fields [17, 18, 16]. At the same time, the classical Rayleigh–Jeans treatment of thermal radiation predicts $u(T) \propto T \omega_{\max}^2$, diverging as the high-frequency cutoff $\omega_{\max} \rightarrow \infty$. The resolution of this “ultraviolet catastrophe” by Planck’s quantization of oscillator energies is one of the historical starting points of quantum theory [13, 14, 15].

In parallel with these developments, classical vortex dynamics has established that incompressible, inviscid fluids support localized vortical structures—thin vortex rings and more general closed vortex loops—with finite energy, impulse, and self-induced translational velocity [19, 20]. In recent work, the present author has summarized standard results for the energy and impulse of thin vortex loops and emphasized their “particle-like” attributes in purely classical Euler flow [22]. A closed filament of circulation Γ and core size a carries a finite kinetic energy $E(\Gamma, R, a)$, an impulse $\mathbf{I}$, and an effective mass $M_{\text{eff}} = \|\mathbf{I}\|/U$, where U is the self-induced velocity along the loop axis. The existence of such finite-energy, localized structures invites the question of how far one can push a classical “vortex-particle” picture within standard continuum mechanics.

A second, complementary observation is that kinetic energy in extended media contributes to inertia and gravitational mass through the relativistic relation $E = mc^2$. While this statement is encoded in the stress–energy tensor of relativistic fluids, explicit examples in simple incompressible configurations remain pedagogically useful. In Ref. [21], an incompressible, inviscid fluid undergoing rigid-body rotation in a finite cylinder was analyzed, and the volume-averaged rotational kinetic energy density $\langle e_{\text{kin}} \rangle$ was related to an effective mass density $\Delta\rho_{\text{eff}} = \langle e_{\text{kin}} \rangle / c^2$ in the nonrelativistic limit. For that geometry one finds the compact result

$$\frac{\Delta\rho_{\text{eff}}}{\rho} = \frac{1}{4} \left(\frac{v_{\text{edge}}}{c} \right)^2, \quad (1.3)$$

where v_{edge} is the tangential speed at the cylinder boundary and ρ is the rest-mass density. This example shows, within standard special relativity, how rotational motion modifies the effective mass density of an incompressible medium at order $(v/c)^2$.

The purpose of the present paper is to combine these two strands—finite-energy vortex loops and effective mass density from rotational kinetic energy—and to explore the thermodynamics of a dilute gas of vortex loops in an incompressible, inviscid fluid. Our analysis is entirely classical and remains within the framework of Eulerian fluid mechanics, special relativity, and textbook statistical mechanics. We show that:

1. A gas of weakly interacting, randomly oriented vortex loops with fixed circulation and core size admits a kinetic description in which the usual ideal-gas equation of state $PV = Nk_B T$ is recovered from the momentum flux, without any modification of Boltzmann’s constant or of the definition of temperature.
2. For an isotropic ensemble of ultra-relativistic or effectively massless excitations built from such loops, the standard relation $P = \frac{1}{3}u$ follows from the three-dimensional structure of the momentum flux tensor, in direct analogy with the radiation case.

3. The mechanical spectrum of admissible loop configurations is effectively bounded from above in frequency by the combination of finite core radius, finite circulation, and a characteristic swirl speed, so that the classical Rayleigh–Jeans integral for the energy density remains finite when evaluated over this mechanically constrained mode set. No modification of Maxwell’s equations or of the relativistic field equations is assumed; the regularization arises from the kinematics of the vortex degrees of freedom.

From a thermodynamic standpoint, the main result is thus that a purely classical, incompressible fluid with vortex-loop microstructure can reproduce the familiar pressure–volume–temperature relations and the $P = \frac{1}{3}u$ relation for an isotropic radiation-like gas, while avoiding the ultraviolet divergence of the naive continuum mode counting. From a conceptual standpoint, the construction provides a concrete mechanical model in which the energy, pressure, and effective mass associated with internal rotational motion can be treated on the same footing as in relativistic continuum mechanics, without invoking any nonstandard dynamics or postulates beyond those already present in classical fluid mechanics and special relativity.

Notation and assumptions

Notation. Throughout the paper we use the following symbols:

Symbol	Meaning
ρ_f	Mass density of the (incompressible, inviscid) fluid
a	Core radius of a circulation loop (vortex-ring core thickness)
R	Ring radius (distance from ring centreline to symmetry axis)
Γ	Circulation of a loop, $\Gamma = \oint \mathbf{v} \cdot d\boldsymbol{\ell}$
M_{eff}	Effective mass of a circulation loop, defined from energy/impulse
v_{max}	Upper bound on tangential speed inside the loop core
ω_{max}	Maximal internal angular frequency, $\omega_{\text{max}} = v_{\text{max}}/a$
u	Energy density of the effective gas or radiation field
P	Pressure of the gas / radiation field
T	Temperature, k_B Boltzmann constant
c	Speed of light in vacuum

Table 1.1: Main symbols used in the circulation-loop model and its thermodynamic description.

Assumptions. The analysis in the main text rests on the following simplifying assumptions:

1. **Incompressible, inviscid background.** The fluid hosting the circulation loops is taken to be homogeneous, incompressible, and inviscid; its density is ρ_f and its mean flow vanishes in the rest frame of the container.
2. **Thin, non-reconnecting loops.** Vorticity is confined to thin, closed circulation loops (vortex rings) with core radius a and ring radius $R \gg a$. Loops are assumed not to reconnect, break, or merge on the timescales of interest.
3. **Dilute gas of loops.** The number density of loops is sufficiently low that mutual induction, deformation, and collisions can be neglected in the first approximation. Each loop can therefore be treated as a quasiparticle with effective mass M_{eff} and centre-of-mass momentum $\mathbf{p}$.
4. **Bounded tangential speed and maximal internal frequency.** The tangential speed of fluid elements within each core obeys $v_\theta \leq v_{\text{max}}$, with $v_{\text{max}} < c$. Together with the finite core radius a , this implies a maximal internal angular frequency $\omega_{\text{max}} = v_{\text{max}}/a$ for rotational motions.

5. **Equipartition / classical limit.** When we assign an average energy $k_B T$ per mode or degree of freedom, we are explicitly in the classical, high-temperature limit of statistical mechanics (equipartition), and we do *not* impose any quantum occupancy constraints on the circulation-loop gas.
6. **Isotropy.** For the radiation-like sector, the ensemble of effective excitations is assumed to have an isotropic momentum distribution in three dimensions, so that angular averages such as $\langle \cos^2 \theta \rangle = 1/3$ apply. In d spatial dimensions this generalizes to $\langle \cos^2 \theta \rangle = 1/d$.

1.2 Vortex-loop gas and kinetic theory

In this section we set up a minimal kinetic model for a dilute gas of vortex loops in an incompressible, inviscid fluid and derive its thermodynamic equation of state. The construction follows standard kinetic theory, with the only nonstandard ingredient being the identification of the microscopic “particles” with finite-energy vortex loops in Euler flow.

1.2.1 Vortex-loop kinematics and effective mass

We consider a homogeneous, incompressible, inviscid fluid of density ρ_f , described by the Euler equations

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho_f} \nabla p, \quad \nabla \cdot \mathbf{v} = 0, \quad (1.4)$$

where $\mathbf{v}(\mathbf{x}, t)$ is the velocity field and $p(\mathbf{x}, t)$ the pressure. Vorticity is defined as

$$\boldsymbol{\omega} = \nabla \times \mathbf{v}. \quad (1.5)$$

We assume that vorticity is confined to thin tubes (vortex filaments) of core radius a , within which the tangential speed is bounded by a characteristic value v_0 , and that the flow outside the cores is irrotational.

Classical vortex dynamics shows that a closed vortex filament of circulation Γ and large-scale radius $R \gg a$ has finite kinetic energy $E(\Gamma, R, a)$, finite impulse $\mathbf{I}$, and a self-induced translational velocity U along its symmetry axis [19, 20, 22]. It is natural to define an effective mass

$$M_{\text{eff}} \equiv \frac{\|\mathbf{I}\|}{U}, \quad (1.6)$$

so that the loop behaves kinematically like a particle of mass M_{eff} moving with velocity U along its axis. In addition to this translational motion, the fluid inside and near the core carries internal rotational kinetic energy. In the nonrelativistic regime, the contribution of this internal energy to the effective mass density can be expressed, following Ref. [21], as

$$\Delta \rho_{\text{eff}} = \frac{\langle e_{\text{kin}} \rangle}{c^2}, \quad (1.7)$$

where $\langle e_{\text{kin}} \rangle$ is the volume-averaged rotational kinetic energy density and c is the speed of light.

In what follows, we treat each vortex loop as a structure with fixed circulation and core size, whose internal degrees of freedom are either frozen or accounted for as a fixed contribution to M_{eff} . The centre-of-mass motion of the loops then admits a standard kinetic description. The explicit thin-ring expressions for impulse, kinetic energy, translational speed, and M_{eff} used in this interpretation are collected in Appendix .1.

1.2.2 Phase space and Hamiltonian

We consider N well-separated vortex loops in a container of volume V . Let $\mathbf{x}_i$ and $\mathbf{p}_i$ denote the centre-of-mass position and momentum of the i -th loop ($i = 1, \dots, N$). In the dilute limit we neglect mutual induction and other interactions between loops, so that the total Hamiltonian factorizes:

$$H = \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2M_{\text{eff}}}. \quad (1.8)$$

The phase space is the usual $6N$ -dimensional space of coordinates and momenta,

$$\Gamma = \{(\mathbf{x}_1, \dots, \mathbf{x}_N; \mathbf{p}_1, \dots, \mathbf{p}_N)\}, \quad (1.9)$$

endowed with the Liouville measure

$$\mathrm{d}\mu(\Gamma) = \prod_{i=1}^N \mathrm{d}^3x_i \mathrm{d}^3p_i. \quad (1.10)$$

Importantly, the incompressibility constraint $\nabla \cdot \mathbf{v} = 0$ is imposed at the level of the underlying Euler flow and affects the detailed form of M_{eff} and the internal energy of each loop. It does not modify the Liouville measure or the Hamiltonian structure for the centre-of-mass coordinates, which remain those of a dilute gas of classical particles.

1.2.3 Canonical ensemble and partition function

We now place the vortex-loop gas in contact with a heat bath at temperature T and describe it within the canonical ensemble. The canonical partition function is

$$Z_N(T, V) = \frac{1}{N! h^{3N}} \int \exp[-\beta H] \mathrm{d}^{3N}p \mathrm{d}^{3N}x, \quad \beta = \frac{1}{k_B T}, \quad (1.11)$$

where h is Planck's constant and k_B Boltzmann's constant. Inserting the Hamiltonian (1.8),

$$Z_N(T, V) = \frac{1}{N! h^{3N}} \prod_{i=1}^N \left[\int_V \mathrm{d}^3x_i \int_{\mathbb{R}^3} \exp\left(-\beta \frac{\mathbf{p}_i^2}{2M_{\text{eff}}}\right) \mathrm{d}^3p_i \right]. \quad (1.12)$$

The position integrals each yield a factor of V , and the momentum integrals are Gaussian:

$$\int_{\mathbb{R}^3} \exp\left(-\beta \frac{\mathbf{p}^2}{2M_{\text{eff}}}\right) \mathrm{d}^3p = (2\pi M_{\text{eff}} k_B T)^{3/2}. \quad (1.13)$$

Hence

$$Z_N(T, V) = \frac{1}{N!} \left[\frac{V}{\lambda_T^3} \right]^N, \quad \lambda_T = \frac{h}{\sqrt{2\pi M_{\text{eff}} k_B T}}, \quad (1.14)$$

where λ_T is the thermal de Broglie wavelength associated with M_{eff} . This is the standard partition function of a classical ideal gas with mass M_{eff} ; the vortex nature of the underlying structures is encoded entirely in the value of M_{eff} and in possible internal-state degeneracies, which we neglect here.

1.2.4 Ideal-gas equation of state

The Helmholtz free energy $F(T, V, N)$ is

$$F(T, V, N) = -k_B T \ln Z_N(T, V) = -k_B T [-\ln N! + N \ln V - 3N \ln \lambda_T]. \quad (1.15)$$

The pressure is obtained in the usual way as

$$P = - \left(\frac{\partial F}{\partial V} \right)_{T,N} = k_B T \left(\frac{\partial \ln Z_N}{\partial V} \right)_{T,N} = k_B T \frac{N}{V}. \quad (1.16)$$

Thus the dilute vortex-loop gas obeys the ideal-gas law

$$PV = Nk_B T, \quad (1.17)$$

with no modification of the equation of state due to the incompressibility of the underlying fluid. The incompressibility affects the value of M_{eff} through the kinetic energy of the core and the surrounding flow, but once M_{eff} is fixed, the centre-of-mass dynamics yields the same P - V - T relation as for point particles.

1.2.5 Isotropic momentum flux and $P = \frac{1}{3}u$

We now turn to the relation between pressure and energy density for an isotropic ensemble of excitations. Consider a gas of particles (here, effective vortex-loop quasiparticles) with energies E and momenta $\mathbf{p}$, moving with speed $v = |\mathbf{v}|$. In kinetic theory, the pressure on a plane normal to the x -axis is given by the momentum flux in the x -direction,

$$P = \frac{1}{V} \left\langle \sum_i p_{x,i} v_{x,i} \right\rangle, \quad (1.18)$$

where the average is over all particles in the volume V and $p_{x,i}$, $v_{x,i}$ are the x -components of the momentum and velocity of particle i . For an isotropic distribution, one can write

$$p_x v_x = (\mathbf{p} \cdot \mathbf{v}) \cos^2 \theta, \quad (1.19)$$

where θ is the angle between $\mathbf{p}$ and the x -axis. In three dimensions the angular average of $\cos^2 \theta$ over the unit sphere is

$$\langle \cos^2 \theta \rangle = \frac{1}{4\pi} \int_0^{2\pi} d\phi \int_0^\pi \cos^2 \theta \sin \theta d\theta = \frac{1}{3}. \quad (1.20)$$

Thus,

$$P = \frac{1}{3V} \left\langle \sum_i \mathbf{p}_i \cdot \mathbf{v}_i \right\rangle. \quad (1.21)$$

In the ultra-relativistic or effectively massless limit, one has $E_i = c\|\mathbf{p}_i\|$ and $\|\mathbf{v}_i\| = c$, so that $\mathbf{p}_i \cdot \mathbf{v}_i = E_i$. The pressure therefore becomes

$$P = \frac{1}{3V} \left\langle \sum_i E_i \right\rangle = \frac{1}{3}u, \quad (1.22)$$

where

$$u = \frac{1}{V} \left\langle \sum_i E_i \right\rangle \quad (1.23)$$

is the energy density. This is the familiar relation for an isotropic gas of massless particles or photons [17, 16]. In the present context, it applies to any regime in which the relevant vortex-loop excitations propagate at speeds close to c and their energy is dominated by translational motion rather than rest energy.

The crucial point is that Eq. (??) follows entirely from the three-dimensional isotropy of the momentum distribution and the kinematics of relativistic particles. The generalization to d

spatial dimensions, yielding $P = u/d$, is derived in Appendix .2; there the factor $1/3$ is seen to be a special case of the purely geometric factor $1/d$.

In summary, a dilute gas of vortex loops in an incompressible, inviscid fluid admits a standard kinetic description in which the ideal-gas law and the radiation-like relation $P = \frac{1}{3}u$ emerge in precisely the same way as for point particles and photons in conventional kinetic theory. The incompressibility of the underlying fluid constrains the internal structure and effective mass of the loops, but does not alter the macroscopic P – V – T relations obtained from their centre-of-mass motion and isotropic momentum flux.

1.3 Mechanically allowed mode set and the classical ultraviolet problem

The derivation of the Rayleigh–Jeans law in a cavity rests on two ingredients: (i) the normal-mode structure of Maxwell’s equations in a finite volume, and (ii) the assumption that each normal mode behaves as an independent harmonic oscillator carrying an average energy $k_B T$ in the classical limit. The number of electromagnetic modes with angular frequency between ω and $\omega + d\omega$ in a volume V is [16]

$$g_{\text{EM}}(\omega) d\omega = \frac{V}{\pi^2 c^3} \omega^2 d\omega, \quad (1.24)$$

so that the classical spectral energy density is

$$u_{\text{RJ}}(\omega, T) d\omega = \frac{\omega^2}{\pi^2 c^3} k_B T d\omega. \quad (1.25)$$

The total energy density $u(T) = \int_0^\infty u_{\text{RJ}}(\omega, T) d\omega$ diverges as ω^3 at the upper limit, giving the well-known ultraviolet catastrophe in the purely classical field picture [14, 15, 13].

In the present framework, the electromagnetic field is regarded as a macroscopic, irrotational manifestation of the motion of an underlying incompressible fluid with circulation-loop microstructure. Maxwell’s equations and their cavity eigenmodes are left intact at the macroscopic level. What changes is the interpretation of the *microstates* underlying a given field configuration: instead of independent field oscillators, the microstates are mechanical configurations of circulation loops. We now show that the kinematics of these loops restricts the set of physically realisable high-frequency modes, effectively regularizing the Rayleigh–Jeans integral without any modification of Maxwell’s equations.

1.3.1 Maximal rotational frequency from core size and tangential speed

As in Sec. 1.2, we consider thin circulation loops (vortex rings) of core radius a embedded in a homogeneous incompressible fluid of density ρ_f . Inside the core, the tangential speed $v_\theta(r)$ of the fluid elements is bounded by a characteristic value v_0 , so that

$$v_\theta(r) \leq v_0 \quad \text{for } 0 \leq r \leq a. \quad (1.26)$$

The associated angular frequency at radius r is

$$\Omega(r) = \frac{v_\theta(r)}{r}, \quad (1.27)$$

so that the maximal angular frequency attainable within the core is bounded by

$$\Omega_{\text{max}} = \max_{0 < r \leq a} \frac{v_\theta(r)}{r} \lesssim \frac{v_0}{a}. \quad (1.28)$$

This bound is purely kinematic: it expresses the fact that no fluid element can circulate around the core with tangential speed exceeding v_0 at a radius smaller than a . In particular, it implies a minimal time scale

$$\tau_{\min} \sim \frac{1}{\Omega_{\max}} \gtrsim \frac{a}{v_0}, \quad (1.29)$$

below which the fluid cannot respond to external forcing or internal excitations without violating the speed bound.

In the present context, we will denote by

$$\omega_{\max} \equiv \Omega_{\max} \quad (1.30)$$

the characteristic maximal angular frequency associated with internal rotational motion in the circulation-loop cores. This frequency provides a mechanical upper bound on how rapidly the substrate can be made to oscillate locally. Any putative field mode with $\omega \gg \omega_{\max}$ would require fluid motions inside the cores with periods shorter than $\tau_{\min}$ and hence tangential speeds above v_0 , which are excluded by construction.

1.3.2 Mechanically realizable field modes

Maxwell's equations in vacuum (or in a homogeneous dielectric) admit harmonic solutions with arbitrarily large wave number k and frequency $\omega = ck$. In the purely field-theoretic picture, nothing prevents one from populating modes with $k \rightarrow \infty$. In the present mechanical picture, however, such field configurations must be realizable as coarse-grained manifestations of the motion of the underlying fluid.

We will not attempt to derive Maxwell's equations from the fluid model. Instead, we make the following conservative, kinematic assumption:

For a given cavity geometry and boundary conditions, we restrict attention to those electromagnetic normal modes whose spatial and temporal variations can be generated by configurations of circulation loops whose internal rotational frequencies do not exceed $\omega_{\max}$.

Operationally, this means that among the continuum of formal cavity eigenmodes, only those with angular frequency ω up to some effective cutoff

$$\omega \lesssim \omega_{\max} \quad (1.31)$$

can actually be excited in thermal equilibrium with the mechanical substrate. Modes with $\omega \gg \omega_{\max}$ exist as mathematical solutions of the field equations, but there is no mechanical microstate in the circulation-loop gas that corresponds to their excitation at finite amplitude; they are therefore excluded from the thermodynamic counting of microstates.

Equivalently, one may say that the *effective* density of thermally accessible modes is modified from Eq. (1.24) to

$$g_{\text{eff}}(\omega) = g_{\text{EM}}(\omega) f\left(\frac{\omega}{\omega_{\max}}\right), \quad (1.32)$$

where $f(x)$ is a dimensionless cutoff function satisfying

$$f(x) \rightarrow 1 \quad (x \ll 1), \quad f(x) \rightarrow 0 \quad (x \gg 1), \quad (1.33)$$

and decaying sufficiently rapidly as $x \rightarrow \infty$ to render the total energy finite. The detailed form of f depends on the microphysics of the circulation-loop gas and on how electromagnetic excitations couple to it; for the present discussion, only the existence of such an f with the stated asymptotics is required.

1.3.3 Modified Rayleigh–Jeans spectrum and finiteness of energy

With the mechanically constrained mode set (1.32), the classical Rayleigh–Jeans spectral energy density becomes

$$u_{\text{eff}}(\omega, T) d\omega = \frac{\omega^2}{\pi^2 c^3} k_B T f\left(\frac{\omega}{\omega_{\text{max}}}\right) d\omega. \quad (1.34)$$

The total energy density is

$$u(T) = \int_0^\infty u_{\text{eff}}(\omega, T) d\omega = \frac{k_B T}{\pi^2 c^3} \int_0^\infty \omega^2 f\left(\frac{\omega}{\omega_{\text{max}}}\right) d\omega. \quad (1.35)$$

By changing variables $x = \omega/\omega_{\text{max}}$, this becomes

$$u(T) = \frac{k_B T}{\pi^2 c^3} \omega_{\text{max}}^3 \int_0^\infty x^2 f(x) dx. \quad (1.36)$$

Under the mild assumption that the integral

$$C_f \equiv \int_0^\infty x^2 f(x) dx \quad (1.37)$$

converges (which follows from the decay of f as $x \rightarrow \infty$), we obtain

$$u(T) = \frac{k_B T}{\pi^2 c^3} \omega_{\text{max}}^3 C_f, \quad (1.38)$$

which is finite for all finite T and scales linearly with temperature in the regime where equipartition holds. The ultraviolet divergence of the naive Rayleigh–Jeans integral is thus removed by the mechanical restriction of the mode set; the formal field-theoretic density of states (1.24) is replaced, for thermodynamic purposes, by the effective density (1.32) that accounts for the finite response time and finite core size of the substrate.

As a simple illustration, consider the sharp-cutoff choice

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & x > 1, \end{cases} \quad (1.39)$$

which encodes the assumption that modes with $\omega \leq \omega_{\text{max}}$ are fully accessible, while those with $\omega > \omega_{\text{max}}$ are inaccessible. In that case,

$$C_f = \int_0^1 x^2 dx = \frac{1}{3}, \quad (1.40)$$

and Eq. (1.38) reduces to

$$u(T) = \frac{k_B T}{3\pi^2 c^3} \omega_{\text{max}}^3. \quad (1.41)$$

This is precisely the Rayleigh–Jeans result with a hard frequency cutoff at ω_{max} , now justified mechanically by the finite core size and maximal tangential speed of the circulation loops. For more realistic, smooth cutoff functions $f(x)$ the prefactor C_f changes, but the qualitative conclusion remains: the energy density is finite and scales as ω_{max}^3 .

1.3.4 Maxwell’s equations and the role of the substrate

It is important to emphasize what has and has not been modified in this construction. At the macroscopic level:

- The form of Maxwell’s equations and their normal-mode solutions in a cavity are left unchanged. The usual mode density (1.24) remains valid as a statement about the spectrum of solutions to the field equations.

- The relations $PV = Nk_B T$ and $P = \frac{1}{3}u$ for the effective quasiparticles (Sec. 1.2) are derived from standard kinetic theory and from the isotropy of the momentum distribution, without any modification of the field equations.

What *does* change is the set of microstates that are admitted when one asks how a given electromagnetic field configuration is realized mechanically. In the usual classical treatment, each normal mode is assigned an independent harmonic-oscillator degree of freedom and hence contributes $k_B T$ to the energy in the classical limit. In the present fluid-based model, the independent microstates are not field amplitudes but configurations of circulation loops obeying the kinematic constraints of incompressible Euler flow. These constraints imply a finite maximal rotational frequency $\omega_{\max}$ and thereby restrict the subset of cavity modes that can be populated in thermal equilibrium with the substrate.

In this sense, the incompressible fluid with circulation-loop microstructure acts as a *mechanical regulator* of the ultraviolet behavior of classical radiation. The Rayleigh–Jeans divergence does not arise, not because the field equations are altered, but because the assumption of an infinite set of independent harmonic oscillators is replaced by a finite-density set of mechanically realizable excitations whose internal time scales cannot be shorter than $1/\omega_{\max}$.

From the thermodynamic point of view, the key result of this section is that the ultraviolet behavior of the classical radiation spectrum is rendered finite once the microstate space is defined in terms of circulation loops with finite core radius and bounded tangential speed. The subsequent sections will explore how this mechanical cutoff scale compares with known microscopic scales in high-energy physics and what, if any, observable consequences might arise in astrophysical or cosmological contexts.

1.4 Microphysical scale and numerical estimates

In Sec. 1.3 we treated the electromagnetic field, in a purely kinematic sense, as being represented by a gas of mechanically admissible modes built from closed circulation loops (thin vortex rings) in an incompressible, inviscid medium. Each loop was assumed to have a finite core radius a and a bounded tangential speed along the core. In this section we translate these mechanical assumptions into (i) an upper bound on the angular frequencies that can be realized by such loops and (ii) the corresponding ultraviolet cutoff for the Rayleigh–Jeans integral. Maxwell’s equations and their normal modes are not modified; the change enters only through the identification of which modes are physically realizable given the microstructure.

1.4.1 Kinematic upper bound on angular frequency

Consider a single closed circulation loop with core radius a embedded in an incompressible, inviscid fluid of density ρ_f . Let r denote the local radius of curvature of the loop centreline, and let $v_\theta(r)$ be the local tangential speed of the circulating fluid inside the core. Locally, the motion is equivalent to rigid-body rotation in a small disc of radius $O(a)$, so that the angular frequency associated with this local rotation is

$$\omega_{\text{loc}}(r) = \frac{v_\theta(r)}{r}. \quad (1.42)$$

We impose two mechanical conditions:

1. *Finite core radius:*

$$r \geq a, \quad (1.43)$$

since the curvature radius of the centreline cannot be smaller than the core radius without leaving the thin-core regime.

2. *Bounded tangential speed:*

$$v_\theta(r) \leq v_{\max}, \quad (1.44)$$

where $v_{\max}$ is a characteristic upper bound, e.g. set by relativistic constraints $v_{\max} < c$ or by the stiffness of the underlying medium.

Combining these two inequalities gives an upper bound on the local rotational frequency,

$$\omega_{\text{loc}}(r) = \frac{v_\theta(r)}{r} \leq \frac{v_{\max}}{a}. \quad (1.45)$$

Since any normal mode of the loop gas is built from superpositions of such local rotational degrees of freedom, it is natural to define the mechanically allowed maximal angular frequency

$$\omega_{\max} := \frac{v_{\max}}{a}. \quad (1.46)$$

Equation (1.46) is a purely kinematic consequence of finite core radius and bounded tangential speed in incompressible Euler flow; no assumption about electromagnetism has been used at this stage.

For later comparison it is helpful to note that a closely related frequency scale already appears as a kinematic reference in rotating-flow benchmarks, where one defines

$$\Omega_0 \equiv \frac{C_e}{r_c}, \quad E_0 = \hbar \Omega_0, \quad (1.47)$$

with C_e a characteristic speed and r_c a microscopic length scale. In the present notation, $\omega_{\max}$ plays the same structural role as Ω_0 if one identifies $v_{\max} \leftrightarrow C_e$ and $a \leftrightarrow r_c$.

1.4.2 Rayleigh–Jeans energy density with a mechanical cutoff

In the standard derivation of the Rayleigh–Jeans law, the electromagnetic field in a cavity is decomposed into normal modes; the number of modes per unit volume with angular frequency between ω and $\omega + d\omega$ is

$$g(\omega) d\omega = \frac{\omega^2}{\pi^2 c^3} d\omega, \quad (1.48)$$

where the factor of 2 for polarization is already included in the prefactor. Equipartition assigns an average energy $k_B T$ per mode, so that the spectral energy density in the classical Rayleigh–Jeans limit reads

$$u_{\text{RJ}}(\omega, T) = g(\omega) k_B T = \frac{k_B T}{\pi^2 c^3} \omega^2. \quad (1.49)$$

If all frequencies $0 \leq \omega < \infty$ are allowed, the total energy density

$$u_{\text{tot}}^{(\text{RJ})}(T) = \int_0^\infty u_{\text{RJ}}(\omega, T) d\omega \quad (1.50)$$

diverges as ω^3 at the upper limit: the classical ultraviolet catastrophe.

In the circulation-loop picture, the *mechanically realizable* modes are restricted by the bound (1.46). If one assumes that only modes with $\omega \leq \omega_{\max}$ can be populated in thermal equilibrium with the substrate, the Rayleigh–Jeans spectrum is automatically cut off:

$$u_{\text{tot}}^{(\text{RJ, cut})}(T) = \int_0^{\omega_{\max}} \frac{k_B T}{\pi^2 c^3} \omega^2 d\omega = \frac{k_B T}{3\pi^2 c^3} \omega_{\max}^3. \quad (1.51)$$

Substituting Eq. (1.46) gives

$$u_{\text{tot}}^{(\text{RJ, cut})}(T) = \frac{k_B T}{3\pi^2 c^3} \left(\frac{v_{\max}}{a} \right)^3. \quad (1.52)$$

A dimensional check confirms that Eq. (1.52) has the correct units: $[k_B T] = \text{J}$ and $[v_{\text{max}}^3/(a^3 c^3)] = \text{m}^{-3}$, so that $[u_{\text{tot}}] = \text{J m}^{-3}$ as required.

More generally, one can write the effect of the mechanical constraint in terms of a dimensionless cutoff function $f(\omega/\omega_{\text{max}})$,

$$u_{\text{eff}}(\omega, T) = \frac{k_B T}{\pi^2 c^3} \omega^2 f\left(\frac{\omega}{\omega_{\text{max}}}\right), \quad (1.53)$$

with $f(x) \rightarrow 1$ for $x \ll 1$ and $f(x) \rightarrow 0$ for $x \gg 1$. The total energy density is then

$$u(T) = \int_0^\infty u_{\text{eff}}(\omega, T) d\omega = \frac{k_B T}{\pi^2 c^3} \omega_{\text{max}}^3 \int_0^\infty x^2 f(x) dx, \quad (1.54)$$

which is finite provided the integral $\int_0^\infty x^2 f(x) dx$ converges. The sharp cutoff used in Eq. (1.51) corresponds to $f(x) = 1$ for $0 \leq x \leq 1$ and $f(x) = 0$ for $x > 1$, giving

$$\int_0^\infty x^2 f(x) dx = \int_0^1 x^2 dx = \frac{1}{3}, \quad (1.55)$$

and reproducing Eq. (1.52).

1.4.3 Numerical scales and relation to a kinematic reference frequency

The expression (1.52) becomes quantitatively meaningful once concrete values for v_{max} and a are specified. In the absence of a fully microscopic model, we keep these parameters symbolic but note that a natural identification is

$$v_{\text{max}} \sim C_e, \quad a \sim r_c, \quad (1.56)$$

where C_e and r_c are a characteristic speed and length scale that already appear in rotating-flow benchmarks, via Eq. (1.47). With this choice,

$$\omega_{\text{max}} \equiv \frac{v_{\text{max}}}{a} \longrightarrow \Omega_0 = \frac{C_e}{r_c}, \quad E_0 = \hbar \Omega_0. \quad (1.57)$$

For illustrative numerical values one may adopt, following Ref. [Iskandarani],

$$\mathbf{v}_\odot \sim 10^6 \text{ m s}^{-1}, \quad r_c \sim 10^{-15} \text{ m}, \quad (1.58)$$

so that

$$\Omega_0 \sim \frac{10^6}{10^{-15}} \text{ s}^{-1} \sim 10^{21} \text{ s}^{-1}, \quad (1.59)$$

and

$$E_0 = \hbar \Omega_0 \sim 10^{-13} \text{ J} \sim 0.5 \text{ MeV}. \quad (1.60)$$

Numerically, using CODATA values, this is of the same order as the electron rest energy $m_e c^2 \approx 0.511 \text{ MeV}$; this numerical proximity is used here only as a reference point for the scale of E_0 .

Substituting $\omega_{\text{max}} = \Omega_0$ into Eq. (1.51), one finds

$$u_{\text{tot}}^{(\text{RJ, cut})}(T) = \frac{k_B T}{3\pi^2 c^3} \Omega_0^3. \quad (1.61)$$

For laboratory blackbodies at $T \sim 10^3\text{--}10^4 \text{ K}$, the characteristic frequencies contributing appreciably to the Planck spectrum lie far below Ω_0 , so the presence of such a mechanical cutoff would have no observable effect on standard thermal-radiation measurements.

1.4.4 Comparison with Planck's law and observational constraints

The exact blackbody spectrum is given by Planck's law,

$$u_{\text{Planck}}(\omega, T) = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{\exp(\hbar\omega/k_B T) - 1}, \quad (1.62)$$

with total energy density

$$u_{\text{tot}}^{(\text{Planck})}(T) = a_R T^4, \quad a_R = \frac{\pi^2 k_B^4}{15 \hbar^3 c^3}. \quad (1.63)$$

By contrast, the cutoff Rayleigh–Jeans result (1.52) scales linearly with T for fixed ω_{max} . No choice of ω_{max} can reconcile this T -scaling with the empirically verified T^4 scaling over a broad range of temperatures. The mechanical cutoff therefore *cannot* replace quantum statistics or Planck's law; it merely shows that, in any mechanical substrate with finite core radius and bounded rotational speeds, the classical equipartition argument does not lead to an ultraviolet divergence at the level of the microstates.

From an observational standpoint, the absence of any high-frequency cutoff in laboratory, nuclear, and astrophysical photon spectra implies a lower bound

$$\omega_{\text{max}} = \frac{v_{\text{max}}}{a} \gg \omega_{\gamma, \text{max}}^{(\text{obs})}, \quad (1.64)$$

where $\omega_{\gamma, \text{max}}^{(\text{obs})}$ is the highest photon frequency reliably measured. This constraint translates into a bound on the ratio v_{max}/a : either the core radius a must be extremely small, or the mechanical speed scale v_{max} extremely large, or both.

In summary, the mechanical mode restriction encoded in Eq. (1.46) yields a natural ultraviolet cutoff $\omega_{\text{max}} = v_{\text{max}}/a$ and a finite Rayleigh–Jeans energy density, without altering Maxwell's equations or the macroscopic equations of state derived in Sec. 1.2. The cutoff does not obviate the need for quantization to recover the Planck spectrum and its T^4 scaling, but it provides a concrete example of how a specific class of underlying rotational microphysics regularizes the classical ultraviolet catastrophe while remaining compatible, in principle, with standard electrodynamics at accessible frequencies.

1.5 Discussion and conclusions

1.5.1 Summary of results

We have considered a classical, incompressible, inviscid fluid in which vorticity is confined to thin circulation loops (vortex rings) of finite core radius a . Within this framework, the main results of the paper are:

1. *Vortex loops as quasiparticles and the ideal-gas law.* Treating closed circulation loops as localized, finite-energy structures with effective mass M_{eff} and centre-of-mass momentum $\mathbf{p}$, we showed that a dilute gas of such loops admits a standard kinetic description. In the canonical ensemble, the partition function reduces to that of a classical ideal gas, and the equation of state

$$PV = Nk_B T \quad (1.65)$$

follows from the usual derivative of the Helmholtz free energy with respect to volume. The incompressibility of the underlying medium affects the value of M_{eff} through the kinetic energy of the core and the surrounding flow but does not modify the macroscopic P – V – T relation.

2. *Isotropic momentum flux and $P = \frac{1}{3}u$.* For an isotropic ensemble of ultra-relativistic or effectively massless excitations built from circulation loops, we recovered the standard relation between pressure and energy density,

$$P = \frac{1}{3}u, \quad (1.66)$$

by computing the momentum flux on a plane and using the three-dimensional angular average $\langle \cos^2 \theta \rangle = 1/3$. This derivation depends only on the isotropy of the momentum distribution and the relativistic relation $E = pc$; it is independent of the detailed microstructure of the underlying fluid.

3. *Mechanical mode restriction and ultraviolet regularization.* By imposing a finite core radius a and a bound $v_{\max}$ on the tangential speed of fluid elements within the core, we obtained a purely kinematic upper bound on the internal rotational frequency,

$$\omega_{\max} = \frac{v_{\max}}{a}. \quad (1.67)$$

If electromagnetic normal modes in a cavity are taken to be in one-to-one correspondence with mechanically realizable excitations of this medium, then only modes with $\omega \leq \omega_{\max}$ can be populated in thermal equilibrium. The classical Rayleigh–Jeans integral for the energy density, when restricted to this mechanically allowed mode set, becomes

$$u_{\text{tot}}^{(\text{RJ}, \text{cut})}(T) = \frac{k_B T}{3\pi^2 c^3} \omega_{\max}^3, \quad (1.68)$$

which is finite for all finite temperatures. More generally, introducing a smooth cutoff function $f(\omega/\omega_{\max})$ yields

$$u(T) = \frac{k_B T}{\pi^2 c^3} \omega_{\max}^3 \int_0^\infty x^2 f(x) dx, \quad (1.69)$$

so that the ultraviolet divergence of the naive Rayleigh–Jeans spectrum is removed once the microstate space is defined in terms of circulation loops with finite core size and bounded rotational speeds.

4. *Consistency with Maxwell’s equations.* Throughout, Maxwell’s equations and their cavity eigenmodes were left unchanged at the macroscopic level. The density of modes $g(\omega) = \omega^2/(\pi^2 c^3)$ remains valid as a statement about the spectrum of solutions to the field equations. The modification enters only through the identification of which modes are physically realizable given the mechanical properties of the underlying medium; the ultraviolet behaviour of the spectrum is thus regularized at the level of microstate counting rather than by altering the field equations.

Taken together, these results demonstrate that a classical incompressible medium with circulation-loop microstructure can reproduce the familiar ideal-gas and radiation equations of state and can regularize the ultraviolet behaviour of the classical Rayleigh–Jeans spectrum, entirely within the framework of Eulerian fluid mechanics, special relativity, and textbook statistical mechanics.

1.5.2 Relation to classical and quantum descriptions of radiation

It is important to distinguish clearly between three levels of description:

1. The *macroscopic field description*, in which the electromagnetic field is governed by Maxwell’s equations in a given background and admits a continuum of normal modes in a cavity, with density $g(\omega) = \omega^2/(\pi^2 c^3)$.

2. The *classical statistical description*, in which each normal mode is treated as an independent harmonic oscillator with average energy $k_B T$ in the high-temperature limit, leading to the Rayleigh–Jeans spectrum and its ultraviolet divergence.
3. The *quantum statistical description*, in which the energy of each mode is quantized in units $\hbar\omega$ and Bose–Einstein statistics yields Planck’s law and the T^4 dependence of the blackbody energy density.

The analysis in this paper modifies only the second of these points. We have shown that if one insists on assigning a classical mechanical substrate to the electromagnetic field—here, an incompressible medium with circulation loops of finite core radius and bounded tangential speeds—then the assumption of an infinite set of independent harmonic oscillators, each carrying $k_B T$, is no longer tenable. Instead, the set of thermally accessible modes is limited by the kinematics of the medium, and the Rayleigh–Jeans integral is rendered finite without altering Maxwell’s equations.

At the same time, the mechanical cutoff introduced by $\omega_{\max} = v_{\max}/a$ does *not* reproduce the correct temperature scaling of the blackbody energy density. For fixed $\omega_{\max}$, the cutoff Rayleigh–Jeans result scales as $u_{\text{tot}}^{(\text{RJ, cut})} \propto T$, whereas Planck’s law yields $u_{\text{tot}}^{(\text{Planck})} \propto T^4$.¹ No choice of $\omega_{\max}$ can reconcile these scalings over a broad range of temperatures. The mechanical regularization discussed here therefore does not remove the need for quantization; it simply illustrates how a particular class of underlying microphysics prevents the classical ultraviolet catastrophe from arising at the level of the substrate.

In this sense, the circulation-loop model should be viewed as a *classical analogue* or toy model: it exhibits, in a purely mechanical setting, how a finite microstructure and a maximum internal frequency can regulate the ultraviolet behaviour of a classical field theory, while leaving intact the macroscopic equations of motion and the standard kinetic derivations of $PV = Nk_B T$ and $P = \frac{1}{3}u$.

1.5.3 Conceptual and pedagogical implications

Even if the actual microscopic origin of electromagnetism lies in quantum fields rather than classical incompressible media, explicit mechanical models can be conceptually useful. The circulation-loop construction offers several such benefits:

- It provides a concrete example in which *internal motion* in an extended medium (here, rotational motion within vortex cores) contributes to effective mass density via $E = mc^2$. This complements textbook discussions of relativistic fluids by exhibiting a specific configuration in which the volume-averaged mass density is shifted by $\Delta\rho_{\text{eff}} = \langle e_{\text{kin}} \rangle / c^2$.
- It illustrates in a tangible way why the naive continuum assumption of infinitely many independent modes can be misleading. In the circulation-loop gas, the irrotational field outside the cores is not a collection of independent oscillators; it is constrained by the finite set of vortex degrees of freedom. The ultraviolet divergence of the Rayleigh–Jeans law is therefore seen as an artefact of overcounting microstates, rather than as an unavoidable pathology of classical physics.
- It connects the statistical mechanics of gases with the dynamics of vortical structures in fluids: the same circulation loops that serve as “particles” in the kinetic theory also carry the rotational energy that underlies the effective mass density and the mechanical cutoff scale.

¹Here and throughout we assume the usual equilibrium photon gas in flat spacetime and neglect chemical potentials.

For teaching purposes, one might imagine a fluid analogue: a tank of water seeded with smoke-ring-like vortices, illuminated and tracked via particle-image velocimetry. As smaller and smaller scales are probed, one eventually reaches the minimal core size and maximal rotation rate of the rings. No matter how finely one tries to decompose the flow into “modes”, there is a physical limit set by the underlying structures. This provides an intuitive picture of how ultraviolet cutoffs can arise from microstructure, independent of any quantum postulate.

1.5.4 Possible extensions and future work

The present work raises several avenues for further investigation:

- *Detailed microphysical models.* We have treated the circulation loops in a schematic way, assuming a fixed core radius a , a fixed speed bound $v_{\max}$, and a simple relation $\omega_{\max} = v_{\max}/a$. A more detailed analysis could derive a , $v_{\max}$, and the effective cutoff function $f(\omega/\omega_{\max})$ from a specific model of the medium, potentially including compressibility, elasticity, or additional internal degrees of freedom.
- *Fluid analogues and laboratory tests.* In classical fluids, wave spectra in the presence of a bath of vortices often exhibit saturation or steepening at high wavenumber, reflecting the finite size of coherent structures. It would be interesting to design controlled experiments in which the energy spectrum of surface or internal waves is measured in a flow populated by vortices of known core size, to test how closely the observed saturation resembles the simple cutoff picture developed here.
- *Coupling to electromagnetism and relativistic field theory.* We have deliberately avoided deriving Maxwell’s equations from the underlying fluid dynamics. One possible extension is to embed the circulation-loop picture in a more general effective field theory, where the fluid degrees of freedom appear as an emergent sector coupled to electromagnetism. This could clarify whether the mechanical cutoff $\omega_{\max}$ can be related to known microscopic scales, such as the electron Compton frequency, without conflicting with precision tests of quantum electrodynamics.
- *Connection to broader microstructural frameworks.* In separate work, circulation loops of the type considered here have been discussed under the name “swirl strings” as part of a broader attempt to represent matter and radiation in terms of structured rotational degrees of freedom in a continuous medium. The present paper can be viewed as the classical, incompressible limit of that wider programme, restricted to thermodynamic and ultraviolet questions. Exploring the full implications of such a framework lies beyond our present scope, but the circulation-loop gas studied here may serve as a useful benchmark sector.

1.5.5 Final remarks

We have shown that a simple, classical model—an incompressible fluid populated by finite-core circulation loops—is sufficient to reproduce the ideal-gas and radiation equations of state and to regularize the classical ultraviolet behaviour of thermal radiation at the level of microstate counting. The ultraviolet catastrophe is thus seen not as an inevitable failure of classical reasoning, but as a consequence of assigning an infinite set of independent degrees of freedom to the field without regard to the mechanical limitations of any underlying medium. While the correct blackbody spectrum and its T^4 scaling undoubtedly require quantum statistics, the circulation-loop model provides a concrete example of how microstructure and finite rotational frequencies can render the classical ultraviolet problem moot, without modifying Maxwell’s equations or the familiar thermodynamic relations that emerge from kinetic theory.

.1 Thin vortex-ring formulas

For completeness we collect here the standard expressions for the impulse, kinetic energy, and translational speed of a thin circular vortex ring in an ideal incompressible fluid, as used in the main text. These formulas go back to Kelvin, Helmholtz, Lamb, and are summarized in modern form by Batchelor and Saffman.

We consider a single circular vortex ring of radius R and core radius a , with $a \ll R$, embedded in a homogeneous incompressible fluid of density ρ_f . The circulation around the core is denoted by Γ .

.1.1 Impulse

The hydrodynamic impulse of an isolated thin-core ring, directed along the symmetry axis, has magnitude

$$I = \rho_f \Gamma \pi R^2. \quad (70)$$

This result is independent of the detailed vorticity distribution inside the core, provided the ring is thin and axisymmetric.

.1.2 Kinetic energy

The kinetic energy associated with the flow field of a thin vortex ring with uniform vorticity in the core is, to leading order in a/R ,

$$E = \frac{1}{2} \rho_f \Gamma^2 R \left(\ln \frac{8R}{a} - \frac{7}{4} \right). \quad (71)$$

The logarithmic term reflects the long-range nature of the velocity field; the constant offset $7/4$ depends weakly on the assumed core model (e.g. uniform vs. Gaussian vorticity).

.1.3 Translational speed

The self-induced translational speed of the ring along its symmetry axis is

$$U = \frac{\Gamma}{4\pi R} \left(\ln \frac{8R}{a} - \frac{1}{4} \right), \quad (72)$$

again valid for $a/R \ll 1$ and for standard core models. Different core profiles change only the constant term inside the parentheses at this order.

.1.4 Effective mass

In the main text we treat the ring as a quasiparticle with effective mass M_{eff} moving at speed U along the symmetry axis, carrying impulse I . It is therefore natural to define the effective mass via $M_{\text{eff}} \equiv I/U$. Using Eqs. (70) and (72), one obtains

$$M_{\text{eff}} = \frac{\rho_f \Gamma \pi R^2}{\frac{\Gamma}{4\pi R} \left(\ln \frac{8R}{a} - \frac{1}{4} \right)} = \frac{4\pi^2 \rho_f R^3}{\ln \frac{8R}{a} - \frac{1}{4}}. \quad (73)$$

For very slender rings, $\ln(8R/a) \gg 1$, the constant term $1/4$ can be neglected and the effective mass simplifies to

$$M_{\text{eff}} \approx \frac{4\pi^2 \rho_f R^3}{\ln(8R/a)}. \quad (74)$$

In the main text, only the scaling $M_{\text{eff}} \propto \rho_f R^3$ and the presence of a weak logarithmic dependence on a/R are used; the precise numerical coefficients play no qualitative role in the thermodynamic arguments.

.2 Pressure and energy density in d dimensions

In the main text we use the relation

$$P = \frac{1}{3}u \quad (75)$$

for an isotropic ensemble of ultra-relativistic or effectively massless excitations in three spatial dimensions. Here we show that in d spatial dimensions the corresponding relation is

$$P = \frac{1}{d}u, \quad (76)$$

so that the factor $1/3$ is a direct consequence of three-dimensional isotropy rather than any special feature of the circulation-loop model.

Consider a gas of non-interacting, massless particles (or effective excitations) in d spatial dimensions, contained in a volume V_d . Let the energy of a particle with momentum $\mathbf{p}$ be

$$E = c|\mathbf{p}|, \quad (77)$$

so that the group velocity is

$$\mathbf{v} = \frac{\partial E}{\partial \mathbf{p}} = c \frac{\mathbf{p}}{|\mathbf{p}|}. \quad (78)$$

We assume an isotropic momentum distribution, so that all directions of $\mathbf{p}$ are equally probable.

.2.1 Energy density

Let $f(\mathbf{p})$ be the occupation number (particles per phase-space volume) in momentum space. The energy density is

$$u = \frac{1}{V_d} \int E(\mathbf{p}) f(\mathbf{p}) d^d p. \quad (79)$$

.2.2 Pressure as momentum flux

The pressure is the normal component of the momentum flux across a plane. Without loss of generality, consider the plane orthogonal to the x -axis. The contribution of a single particle with momentum $\mathbf{p}$ and velocity $\mathbf{v}$ to the x -component of the momentum flux is $p_x v_x / V_d$. Summing over all particles,

$$P = \frac{1}{V_d} \int p_x v_x f(\mathbf{p}) d^d p. \quad (80)$$

For massless particles, $v_x = c p_x / |\mathbf{p}|$, so

$$p_x v_x = c \frac{p_x^2}{|\mathbf{p}|}. \quad (81)$$

It is convenient to write $\mathbf{p}$ in spherical coordinates in momentum space: $|\mathbf{p}| = p$, and let θ be the angle between $\mathbf{p}$ and the x -axis. Then $p_x = p \cos \theta$ and

$$p_x v_x = c \frac{p^2 \cos^2 \theta}{p} = c p \cos^2 \theta. \quad (82)$$

Substituting into Eq. (80),

$$P = \frac{c}{V_d} \int p \cos^2 \theta f(\mathbf{p}) d^d p. \quad (83)$$

.2.3 Isotropic angular average

For an isotropic distribution, the angular dependence factorizes:

$$f(\mathbf{p}) = f(p), \quad (84)$$

and the average of $\cos^2 \theta$ over the unit sphere in d dimensions is

$$\langle \cos^2 \theta \rangle = \frac{1}{d}. \quad (85)$$

This is a standard result: by symmetry, $\langle p_x^2 \rangle = \langle p_y^2 \rangle = \dots = \langle p_d^2 \rangle$, and $\sum_{i=1}^d \langle p_i^2 \rangle = \langle p^2 \rangle$, hence $\langle p_x^2 \rangle = \langle p^2 \rangle / d$ and $\langle \cos^2 \theta \rangle = \langle p_x^2 / p^2 \rangle = 1/d$.

Using this in Eq. (83), we can write

$$P = \frac{c}{V_d} \langle \cos^2 \theta \rangle \int p f(p) d^d p = \frac{c}{d V_d} \int p f(p) d^d p. \quad (86)$$

Since $E = cp$, the integral $\int p f(p) d^d p = (1/c) \int E f(p) d^d p$, and comparing with Eq. (79) we obtain

$$u = \frac{1}{V_d} \int E(\mathbf{p}) f(\mathbf{p}) d^d p = \frac{c}{V_d} \int p f(p) d^d p. \quad (87)$$

Therefore Eq. (86) becomes

$$P = \frac{1}{d} u. \quad (88)$$

Specializing to $d = 3$ recovers the familiar relation $P = \frac{1}{3}u$ used in the main text. The result is completely general for massless particles with isotropic momentum distributions and does not depend on the details of the circulation-loop model; it is a kinematic statement about momentum flux in d -dimensional space.

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Appendix A

SST-24 — Thermodynamics framework

Thermodynamics of a Circulation-Loop Gas in an Incompressible Fluid and the Classical Ultraviolet Problem

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Abstract

We develop a comprehensive thermodynamic formulation of Swirl–String Theory (SST), in which the physical vacuum is modeled as a frictionless, incompressible swirl condensate, and all matter arises as topologically stabilized vortex filaments (“swirl strings”). Building on the quantum–thermodynamic isomorphism of Abe & Okuyama, we demonstrate that hydrogenic structure, particle masses, vacuum fluctuations, and interaction lifetimes can be reinterpreted as thermodynamic processes involving the swelling, compression, and mode–excitation of vortex cores.

We define SST Work as the mechanical energy required to deform the vortex core radius against the surrounding swirl pressure, and SST Heat as the energy redistributed among Kelvin modes and topological phase channels. Applying this framework to hydrogen, we show that the Bohr radius is not a probabilistic orbital shell but a thermodynamic equilibrium surface where centrifugal swirl pressure balances vacuum tension.

We reinterpret the Golden Layer mass hierarchy as a discrete thermodynamic scaling law governed by the golden ratio, emerging from log–periodic structure in the swirl energy density.

The Unruh Echo is shown to arise from a two–stage thermodynamic response: a 0.1 ns vorticity burst followed by a delayed electromagnetic transduction pulse around 30 ns.

Finally, we derive partition functions, heat capacities, and provide numerical evaluation for a simplified Golden ladder.

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A.1 Introduction: The Hydrodynamic Necessity

A.1.1 Ontological Divergence in Modern Physics

General Relativity models gravity as spacetime curvature, while Quantum Field Theory models matter as excitations of an operator-valued vacuum. These frameworks are empirically successful yet mathematically incompatible.

Swirl-String Theory (SST) replaces both ontologies with a physical one: the vacuum is a real fluid with density, pressure, vorticity, and circulation. Matter is reinterpreted as knotted vortex filaments of this medium. Mass, time dilation, charge, binding energy and field structure all emerge as hydrodynamic responses.

A.1.2 The Thermodynamic Hypothesis

SST has precise formulations for kinematics, field analogues, knot mass spectra, and hydrogen structure, but lacks a unifying thermodynamic interpretation. Abe & Okuyama showed that the Schrödinger equation can be reformulated as a thermodynamic equation of state if one assigns Shannon entropy to the quantum probabilities and imposes the Clausius equality.¹ Their mapping between probability flows and thermodynamic heat is the missing bridge for SST.

A.1.3 Objectives of this Framework

Our aims:

1. Define the swirl string as a thermodynamic system embedded in a reservoir.
2. Identify its thermodynamic variables: core radius, Kelvin modes, topology.
3. Define SST Work as geometric deformation of r_c .
4. Define SST Heat as Kelvin-mode or R/T-phase redistribution.
5. Derive the SST equation of state from Euler-Bernoulli and circulation laws.
6. Map Abe-Okuyama adiabatic/isothermal processes to SST swelling dynamics.
7. Apply the framework to hydrogen, Golden Layers, Unruh Echo and partition functions.

A.2 The Hydrodynamic Substrate: Axioms and Constants

A.2.1 The Primitive Triad

SST is based on three primitive medium parameters:

- Circulation quantum:

$$\Gamma_0 \approx 6.4 \times 10^3 \text{ m}^2/\text{s}. \quad (\text{A.1})$$

- Core radius:

$$r_c \approx 1.41 \times 10^{-15} \text{ m}. \quad (\text{A.2})$$

- Effective fluid density:

$$\rho_f \approx 7.0 \times 10^{-7} \text{ kg/m}^3. \quad (\text{A.3})$$

¹S. Abe and S. Okuyama, “Similarity between quantum mechanics and thermodynamics: Entropy, temperature, and Carnot cycle,” *Phys. Rev. E* **83**, 021121 (2011).

Via circulation conservation $\Gamma = vr$, this yields the swirl speed scale

$$\mathbf{v}_\odot = \frac{\Gamma_0}{2\pi r_c} \approx 1.09 \times 10^6 \text{ m/s.} \quad (\text{A.4})$$

This is the characteristic “sound” speed of swirl excitations.

A.2.2 The Mass Kernel as Equation of State

SST mass arises from the hydrodynamic energy of the vortex:

$$M(T) = \Lambda_0 \mathcal{I}_M(K(T)) L_{\text{tot}}(T), \quad (\text{A.5})$$

with $\Lambda_0 \propto \rho_f \mathbf{v}_\odot^2 r_c^3$, a topological invariant $\mathcal{I}_M$, and total ropelength L_{tot} . Thus rest mass is interpreted as stored adiabatic work.

A.2.3 The Swirl Clock

Time dilation follows:

$$S_t = \sqrt{1 - \frac{v^2}{c^2}}. \quad (\text{A.6})$$

Since $v \sim 1/r$, smaller radii produce larger speeds and slower clocks. This couples energy density directly to local temporal flow.

A.3 The Quantum–Thermodynamic Isomorphism (Abe–Okuyama)

A.3.1 Particle in a Box vs. Vortex in a Core

Quantum energies scale as:

$$E_n(L) \propto \frac{1}{L^2}. \quad (\text{A.7})$$

Vortex energies scale as:

$$E(r_c) \propto \frac{1}{r_c^2}. \quad (\text{A.8})$$

Hence the mappings:

$$L \leftrightarrow r_c, \quad |u_n\rangle \leftrightarrow \text{Kelvin modes.} \quad (\text{A.9})$$

A.3.2 Mapping of Variables

Abe–Okuyama	SST Equivalent	Interpretation
Well width L	Core radius r_c	Confinement scale
Probabilities p_n	Kelvin mode weights	Microstates
Heat	$\sum E_n dp_n$	Mode excitation
Work	$\sum p_n dE_n$	Core deformation
Entropy	Shannon entropy	Topological/Modal complexity

A.3.3 First and Second Laws

Quantum variation:

$$dE = \sum_n E_n dp_n + \sum_n p_n dE_n. \quad (\text{A.10})$$

Thus:

$$\delta Q = \sum_n E_n dp_n, \quad \delta W = \sum_n p_n dE_n. \quad (\text{A.11})$$

In SST:

- Heat = redistribution of Kelvin modes.
- Work = deformation of r_c .

A.3.4 Summary comparison: Abe–Okuyama vs. SST

The Abe–Okuyama (AO) framework treats thermodynamics as an information–theoretic structure defined on a fixed Hilbert space. Temperature appears as a Lagrange multiplier enforcing a constraint on the Shannon entropy of expansion probabilities, and no physical medium is introduced. In Swirl–String Theory (SST), by contrast, thermodynamics is mechanical: temperature, heat, and work are properties of a continuous swirl condensate and its topological defects.

In AO, the microscopic degrees of freedom are the amplitudes $\{c_n\}$ or probabilities $\{p_n\}$ in an energy eigenbasis $\{|u_n\rangle\}$; the geometry of space does not deform when the system is heated or cooled. Heat is associated with changes dp_n at fixed spectrum $\{E_n\}$, while work is associated with changes in the spectrum dE_n at fixed $\{p_n\}$ due to external parameter variation (e.g. moving the walls of a box of width L).

In SST, the microscopic degree of freedom is the radius of a swirl core and its surrounding equilibrium boundary. The relevant strain variable is the geometric deformation of the core/orbital radius away from equilibrium, encoded in the swirl temperature T_{swirl} . Heat corresponds to the excitation and damping of Kelvin modes and topological channels at fixed core radius, while work is the mechanical swelling or compression of the core against the ambient swirl pressure.

These different ontologies lead to distinct equations of state for the energy–temperature relationship. SST predicts a quadratic low–temperature law for the swelling mode,

$$E_{\text{SST}}(T_{\text{swirl}}) \propto T_{\text{swirl}}^2, \quad C_V^{\text{SST}}(T_{\text{swirl}}) \equiv \frac{dE_{\text{SST}}}{dT_{\text{swirl}}} \propto T_{\text{swirl}}, \quad (\text{A.12})$$

whereas a gapped AO two–level truncation yields a Schottky–type behavior

$$C_V^{\text{AO}}(T) \propto \left(\frac{\Delta E}{k_B T}\right)^2 \exp\left(-\frac{\Delta E}{k_B T}\right), \quad (\text{A.13})$$

vanishing exponentially as $T \rightarrow 0$ and approaching a constant high–temperature limit. The qualitative contrast $C_V^{\text{SST}} \sim T_{\text{swirl}}$ vs. $C_V^{\text{AO}} \sim \exp(-\Delta E/k_B T)$ provides a clear empirical discriminator between the two vacua, and will be used below as the basis for low–temperature and log–periodic heat capacity tests.

A.4 Entropy versus Geometric Swelling in the Swirl Condensate

In the preceding sections we have established: (i) that temperature in Swirl–String Theory (SST) is identified with the radial strain of equilibrium boundaries, and (ii) that the Abe–Okuyama mapping furnishes a thermodynamic decomposition of the vortex energy in terms of Kelvin modes. In this section we clarify the distinction between *geometric swelling* and *entropy*, and then outline two applications: nuclear entropy modulation and cosmological thermodynamics.

A.4.1 Abe–Okuyama decomposition and the SST first law

For a vortex filament with discrete Kelvin modes labeled by n , let E_n denote the mode energies and p_n the corresponding occupation probabilities. The total energy may be written as

$$E = \sum_n p_n E_n. \quad (\text{A.14})$$

Feature	Abe–Okuyama (AO)	Swirl–String Theory (SST)
Ontology	Pure states in Hilbert space; no explicit medium	Physical swirl condensate with density, pressure, vorticity; particles are knotted defects
Temperature	Lagrange multiplier enforcing the Clausius relation for Shannon entropy	Geometric strain of an equilibrium boundary; $T_{\text{swirl}} = \Theta \epsilon$ encodes radial swelling
Heat Q	Energy change from dp_n at fixed spectrum $\{E_n\}$	Kelvin–mode and topological excitation at fixed core radius r_c
Work W	Energy change from dE_n at fixed $\{p_n\}$ via external parameter variation (e.g. box width L)	Mechanical swelling/compression of the core and orbital envelope against vacuum swirl pressure
Equation of state	Low- T energy changes governed by gapped spectrum; two-level truncation gives Schottky behavior	Swelling mode obeys $E_{\text{SST}} \propto T_{\text{swirl}}^2$ for small strain, defining an elastic equation of state
Heat capacity C_V	$C_V^{\text{AO}}(T) \sim (\Delta E/k_B T)^2 e^{-\Delta E/k_B T}$ at low T ; approaches a constant at high T	$C_V^{\text{SST}}(T_{\text{swirl}}) \propto T_{\text{swirl}}$ at low T_{swirl} , with possible log–periodic oscillations from Golden–Layer structure
Entropy	Shannon entropy of probability distribution $\{p_n\}$ in a fixed basis	Geometric/phase–space measure of fluctuating boundaries plus discrete Golden layering of the vacuum
Stability	Preventing collapse relies on external potentials and the uncertainty principle	Stability arises from pressure balance of swirl flow, core tension, and Golden–filtered vacuum enthalpy
Time	External parameter t independent of temperature or strain	Internal Swirl Clock $S_t(t) = \sqrt{1 - v^2/c^2}$ tied to local swirl speed; heating/swelling changes proper time
Model motif	Particle–in–a–box with externally imposed boundaries	Self–confined harmonic swelling of a knotted ring; confinement and mass come from the same enthalpy functional

Table A.1: Structural comparison between the information–theoretic thermodynamics of Abe–Okuyama and the hydrodynamic thermodynamics of Swirl–String Theory. AO assigns thermodynamic meaning to probability flows in a fixed Hilbert space, while SST interprets thermodynamic variables as geometric properties of a swirl condensate and its defects. The different ontologies lead to distinct predictions for low–temperature heat capacity, stability mechanisms, and the coupling between temperature and proper time.

A general variation decomposes as

$$dE = \sum_n E_n dp_n + \sum_n p_n dE_n. \quad (\text{A.15})$$

Following the quantum–thermodynamic isomorphism, we identify

$$\delta Q \equiv \sum_n E_n dp_n, \quad \delta W \equiv \sum_n p_n dE_n, \quad (\text{A.16})$$

so that the first law takes the standard form

$$dE = \delta Q + \delta W. \quad (\text{A.17})$$

In SST the geometric confinement is characterized by a core radius r_c and (for bound states such as hydrogen) an orbital radius R . The spectrum E_n is primarily a function of the confinement scale, $E_n = E_n(r_c, R)$, while the probabilities p_n encode how the available energy is distributed among Kelvin modes. Thus:

- δW corresponds to *geometric deformation*, i.e. changes in r_c and R at fixed $\{p_n\}$. In the continuum limit this coincides with $P dV$ work against the vacuum pressure of the swirl condensate.
- δQ corresponds to *mode excitation*, i.e. changes in $\{p_n\}$ at fixed geometry. This is the primary carrier of heat in the swirl sector.

This separation is independent of any specific equation of state and follows directly from the Abe–Okuyama decomposition.

A.4.2 Temperature as swelling; entropy as complexity

For bound states with two intrinsic length scales—a core radius r_c and an orbital radius a_0 —the SST Canon defines the *swirl temperature* T_{swirl} as a measure of radial strain. Introducing the dimensionless strains

$$\epsilon_c = \frac{r - r_c}{r_c}, \quad \epsilon_o = \frac{R - a_0}{a_0}, \quad (\text{A.18})$$

and using the canonical coupling between r_c and a_0 , small deformations satisfy $\epsilon_c \simeq \epsilon_o \equiv \epsilon$. Temperature is then defined by

$$T_{\text{swirl}} \equiv \Theta \epsilon \simeq \Theta \frac{R - a_0}{a_0}, \quad (\text{A.19})$$

where Θ is a stiffness-dependent scale with units of Kelvin. Thus:

$$T_{\text{swirl}} = 0 \iff R = a_0, \quad r = r_c \quad (\text{ground-state geometry}), \quad (\text{A.20})$$

while $T_{\text{swirl}} > 0$ corresponds to a swollen orbital envelope and core.

By contrast, the *Kelvin entropy* S_K depends only on the mode probabilities $\{p_n\}$:

$$S_K = -k_B \sum_n p_n \ln p_n. \quad (\text{A.21})$$

A perfectly smooth filament with all energy in the fundamental geometry ($p_0 = 1$) has $S_K = 0$, regardless of its radius. A filament carrying a broad, disordered spectrum of Kelvin waves has large S_K even if its mean radius is unchanged.

In addition there is a *topological* contribution S_{top} associated with the knot type K . In SST the mass functional includes a Golden suppression factor of the form

$$w(K) \propto \phi^{-g(K)}, \quad (\text{A.22})$$

with ϕ the Golden Ratio and $g(K)$ a topological index (e.g. a genus or complexity measure). This can be interpreted as a Boltzmann-like weight $w(K) \propto \exp[-S_{\text{top}}(K)/k_B]$ with

$$S_{\text{top}}(K) \propto g(K) \ln \phi. \quad (\text{A.23})$$

The total entropy of a bound swirl string can therefore be written as

$$S = S_K + S_{\text{top}}. \quad (\text{A.24})$$

It is a measure of *flow complexity*—modal and topological—and not of the mere volume of the irrotational envelope.

We can now answer a natural question: *Is entropy simply the swelling of the irrotational flow?* Within SST the answer is no:

- Swelling of the irrotational region changes R and thus V , and appears primarily in δW ; it is the geometric component of the first law and constitutes work against the confining vacuum pressure.

- Entropy measures the number and complexity of accessible microstates, encoded in the Kelvin mode distribution and knot topology, and appears in δQ and in the statistical weights of topological sectors.

Swelling and entropy are correlated in realistic processes—since mode heating back-reacts on geometry—but they are distinct thermodynamic coordinates.

A.4.3 Thermodynamic application I: nuclear entropy modulation

At nuclear scales SST models nucleons as knotted swirl strings embedded in a tightly coupled lattice. A bound nucleus is represented as a linked network of such knots, with binding energy arising from constructive interference of their swirl fields.

A nucleon knot possesses a characteristic swirl resonance frequency set by the canonical swirl speed and core radius,

$$\Omega_0 \sim \frac{\|\mathbf{v}_\mathcal{O}\|}{r_c}. \quad (\text{A.25})$$

With the canonical values $\|\mathbf{v}_\mathcal{O}\| \approx 1.09 \times 10^6 \text{ m/s}$ and $r_c \approx 1.41 \times 10^{-15} \text{ m}$ one finds

$$\Omega_0 \sim 8 \times 10^{20} \text{ s}^{-1}, \quad (\text{A.26})$$

i.e. a natural frequency scale of order 10^{21} s^{-1} .

In this picture, an external field structured at or near Ω_0 can, in principle, pump energy preferentially into Kelvin modes of the nucleon knots without substantially changing their mean radius on short time scales. From the thermodynamic viewpoint this is a process with $\delta Q > 0$ and initially small δW , increasing S_K while keeping T_{swirl} approximately fixed.

We may distinguish two qualitative phases:

- A *T-phase* (“tangible”): Kelvin entropy is low, the knot is localized, and its effective interaction range is short. Overlap with neighboring knots is suppressed by the usual Coulomb-like repulsion.
- An *R-phase* (“radiative”): Kelvin entropy is high, the knot becomes delocalized over a larger effective support, and the swirl field extends more coherently into the surrounding lattice.

Driving a nucleon from T-phase toward R-phase via resonant Kelvin excitation increases its thermodynamic overlap with neighboring nucleons. In an SST interpretation, fusion can then proceed by *re-knotting* of overlapping swirl strings, rather than by brute-force kinetic overcoming of the Coulomb barrier. This suggests a thermodynamic “backdoor” to fusion based on entropy modulation rather than high thermal velocities.

At present this is a conjectural application of the SST thermodynamic framework. A quantitative model would require: (i) an explicit Kelvin-mode spectrum for nuclear knots; (ii) a coupling model for structured external fields; and (iii) a calculation of fusion rates as a function of S_K and S_{top} . We include it here as a concrete example of how the distinction between swelling and entropy may be exploited in principle.

A.4.4 Thermodynamic application II: cosmological backreaction

On cosmological scales the swirl condensate fills space with a statistical distribution of vorticity. Let $\langle \omega^2 \rangle$ denote an appropriately coarse-grained variance of the vorticity field. In analogy with backreaction approaches to inhomogeneous cosmology,² one may introduce a kinematical backreaction scalar $\mathcal{Q}_\mathcal{D}$ that depends on the variance of expansion, shear, and vorticity over a domain $\mathcal{D}$.

²See e.g. T. Buchert, *Gen. Relativ. Gravit.* **32**, 105–125 (2000).

Within SST it is natural to regard part of the effective cosmological term $\Lambda_{\text{SST}}(t)$ as a hydrodynamic contribution of the form

$$\Lambda_{\text{SST}}(t) \sim -\frac{1}{2} \mathcal{Q}_{\mathcal{D}}[\langle \omega^2 \rangle], \quad (\text{A.27})$$

so that the large-scale acceleration of the universe is interpreted as a relaxation of swirl inhomogeneities. In the early universe, intense vorticity and small-scale structure correspond to a low-entropy but high-pressure configuration of the swirl condensate. As the universe expands, Kelvin modes and larger-scale flow structures are excited, increasing the entropy S while redistributing internal energy. The resulting decay of $\langle \omega^2 \rangle$ feeds back on $\Lambda_{\text{SST}}(t)$, altering the effective pressure that drives cosmic expansion.

This cosmological sketch is intentionally minimal. Its purpose here is to illustrate how the thermodynamic quantities defined above—temperature as swelling, entropy as modal and topological complexity, and pressure as swirl tension—extend consistently from atomic to cosmological scales within a single hydrodynamic ontology. A full cosmological model would require coupling the SST energy–momentum tensor to an effective large-scale metric and deriving observational signatures (e.g. in the CMB or structure growth) of the log-periodic Golden-layer thermodynamics.

A.5 Multi-Scale Thermodynamics of the Swirl Condensate

A.5.1 The Swirl String as Thermodynamic System

The swirl string is immersed in the swirl condensate reservoir. Thermodynamic variables:

- r_c : geometric confinement coordinate.
- $\{p_n\}$: Kelvin–mode populations.
- Topological sector K : conserved under adiabatic evolution.

A.5.2 SST Work: Core Deformation

Euler’s radial momentum equation gives:

$$\frac{dp}{dr} = \rho_f \frac{v_\theta^2}{r}, \quad v_\theta = \frac{\Gamma}{2\pi r}, \quad (\text{A.28})$$

with Γ the circulation. Integrating:

$$p(r) = p_\infty - \frac{1}{2} \rho_f v_\theta^2. \quad (\text{A.29})$$

Thus the compressive tension at r_c scales as:

$$p(r_c) \propto \frac{1}{r_c^2}. \quad (\text{A.30})$$

Work associated with changing r_c :

$$\delta W_{\text{SST}} = \left(\frac{\partial H}{\partial r_c} \right) dr_c \propto \frac{dr_c}{r_c^3}. \quad (\text{A.31})$$

A.5.3 SST Heat: Kelvin Mode Excitation

At fixed geometry:

$$\delta Q_{\text{SST}} = \sum_n E_n(r_c) dp_n. \quad (\text{A.32})$$

Heat corresponds to:

- mode activation,
- Kelvin–helix excitations,
- R/T-phase reconfigurations.

A.5.4 Adiabatic vs. Isothermal Swelling

Adiabatic (entropy fixed, topology protected):

$$fr_c^3 = \text{const.} \quad (\text{A.33})$$

Isothermal (vacuum temperature fixed):

$$fr_c = \text{const.} \quad (\text{A.34})$$

Hydrogen formation proceeds isothermally; rest mass originates adiabatically.

A.6 Application I: Thermodynamic Origin of Hydrogen Structure

A.6.1 Two-Scale Geometry: $r_c \rightarrow a_0$

A swirl string representing the electron possesses two relevant geometric scales:

$$r_c \approx 1.41 \times 10^{-15} \text{ m}, \quad a_0 \approx 5.29 \times 10^{-11} \text{ m}. \quad (\text{A.35})$$

The SST Canon provides the scaling relation:

$$a_0 = \frac{c^2}{2\mathbf{v}_0^2} r_c, \quad (\text{A.36})$$

which quantitatively reproduces the Bohr radius. Thus the electronic vortex undergoes a thermodynamic expansion from r_c to a_0 during atom formation.

A.6.2 Hydrogen Formation as an Isothermal Expansion

During hydrogen binding, the electron vortex moves inward toward the protonic swirl–Coulomb potential. The relevant force is a compressive hydrodynamic pressure:

$$P(r) = P_\infty - \frac{1}{2}\rho_f v_\theta^2(r), \quad v_\theta = \frac{\Gamma_0}{2\pi r}. \quad (\text{A.37})$$

As r decreases, v_θ increases, lowering the internal pressure at the core boundary and generating a net inward force.

Thermodynamically, this corresponds to *mechanical work* performed on the swirl string. However, the system must remain in equilibrium with the reservoir’s swirl fluctuations. Thus, binding proceeds through an isothermal process.

The equilibrium point occurs when:

$$\frac{dF}{dr} = 0, \quad F = E - T_{\text{sw}}S, \quad (\text{A.38})$$

with S the topological/Kelvin entropy and T_{sw} the swirl reservoir temperature. This yields precisely $r = a_0$ as the stable radius.

A.6.3 Clock Rate Difference: Cold vs. Hot States

The swirl clock:

$$S_t(r) = \sqrt{1 - \frac{v_\theta^2(r)}{c^2}} \quad (\text{A.39})$$

implies:

- Small r : large swirl speed, strong time dilation.
- Large r : weaker swirl, faster internal clock.

Thus the hydrogen ground state (at a_0) is not only an energy minimum, but a *slow-clock minimum*. Excited states have larger radii and therefore experience less time dilation, accelerating spontaneous decay.

A.6.4 Hydrogen as Thermodynamic Boundary Balance

Hydrogen is stable because:

$$\frac{1}{2}\rho_f v_\theta^2(a_0) \quad \text{balances} \quad P_{\text{vac}}, \quad (\text{A.40})$$

and any small displacement increases free energy. Thus atomic structure is fundamentally a thermodynamic equilibrium of swirl pressure and vacuum tension.

A.7 Application II: Thermodynamics of the Golden Layer

A.7.1 The Golden Ratio as Critical Thermodynamic Exponent

The SST mass functional includes a Golden suppression factor:

$$M(K) \propto \phi^{-g(K)} n(K)^{-1/\phi}, \quad (\text{A.41})$$

with g the genus and n the number of components. This resembles a Boltzmann factor:

$$e^{-\Delta E/kT} \leftrightarrow \phi^{-g}. \quad (\text{A.42})$$

Thus high-genus topologies are *thermodynamically suppressed* in the swirl condensate. Mass emerges from the competition between:

- geometric stretching energy (ropelength),
- vacuum displacement energy,
- topological entropy measured by Golden scaling.

A.7.2 Golden Potential and Log-Periodic Structure

Define the Golden potential for swirl energy density ρ_E :

$$V_\phi(\rho_E) = \Lambda^4 \left[1 - \cos \left(\frac{2\pi}{\ln \phi} \ln \frac{\rho_E}{\rho_E^*} \right) \right]. \quad (\text{A.43})$$

Since $\rho_E \propto 1/r^2$, the core stability inherits a log-periodic structure:

$$r_{n+1} = r_n \phi. \quad (\text{A.44})$$

These are the “Golden Layers”—thermodynamic attractors of the vortex core.

A.7.3 Fractal Heat Capacity

A log-periodic system has a heat capacity of the form:

$$C_V(T) = C_0 \left[1 + A \cos \left(\frac{2\pi}{\ln \phi} \ln \frac{T}{T_*} + \delta \right) \right]. \quad (\text{A.45})$$

This predicts *oscillatory heat capacity* vs. $\ln T$ —a key experimental signature of Golden scaling.³

A.8 Application III: The Unruh Echo as Thermodynamic Response

A.8.1 Two-Vacuum SST Interpretation

Standard Unruh radiation arises from coupling to the electromagnetic vacuum with speed c .⁴ SST introduces a second vacuum sector: the swirl medium with characteristic velocity $\mathbf{v}_\odot$.

Accelerated motion induces:

$$T_{\text{sw}}(a) \propto a, \quad (\text{A.46})$$

a thermal excitation of the swirl condensate.

A.8.2 Two-Stage Thermodynamic Pulse

Experiments observing a delayed EM flash after mechanical excitation show:

- A primary burst at ~ 0.1 ns (swirl-sector excitation, pure vorticity).
- A secondary “echo” at ~ 30 ns (EM-sector transduction due to impedance mismatch).

This corresponds schematically to:

$$\delta Q_{\text{sw}}(t=0.1\text{ns}) \rightarrow \delta W_{\text{EM}}(t=30\text{ns}), \quad (\text{A.47})$$

where Kelvin-mode heat is converted into EM work via boundary coupling.

A.8.3 Thermodynamic Explanation

1. Acceleration stretches vorticity (raising local swirl temperature).
2. This excites Kelvin modes (SST Heat).
3. Kelvin modes interact with charges or surfaces.
4. Part of that energy is re-radiated electromagnetically.

Thus the Unruh effect is a *heat*→*work transduction* problem between two coupled vacuum sectors.

³See D. Sornette, Phys. Rep. **297**, 239–270 (1998).

⁴W. G. Unruh, Phys. Rev. D **14**, 870–892 (1976).

A.9 Application IV: Partition Function and Multi-Scale Heat Capacity

A.9.1 Golden Ladder Spectrum

For a simplified proton ladder:

$$E_n = E_0 \phi^n. \quad (\text{A.48})$$

The partition function:

$$Z(\beta) = \sum_{n=0}^N e^{-\beta E_0 \phi^n}. \quad (\text{A.49})$$

Internal energy:

$$U = -\frac{\partial}{\partial \beta} \ln Z. \quad (\text{A.50})$$

Heat capacity:

$$C_V = \frac{\partial U}{\partial T} = k_B \beta^2 (\langle E^2 \rangle - \langle E \rangle^2). \quad (\text{A.51})$$

A.9.2 Numerical Behavior

For $E_0 \sim 300$ MeV and $N \sim 10$, one finds:

- A smooth baseline increase of $C_V(T)$,
- Small log-periodic modulations of amplitude $\sim 1\text{--}5\%$,
- No divergence, consistent with finite-level truncation.

Full fractal oscillations require an extended Golden ladder.

A.9.3 Physical Interpretation

A rising heat capacity with subtle log-oscillations implies:

- The swirl condensate stores thermal energy in discrete topological channels,
- Kelvin-mode activation thresholds are distributed $\propto \phi^n$,
- Proton structure is thermodynamically multi-layered.

A.10 Outlook

This thermodynamic framework suggests several research directions:

1. **Kelvin-mode spectroscopy:** detect logarithmic spacing in excitation energies.
2. **Unruh Echo experiments:** test the two-stage (0.1 ns vs. 30 ns) response predicted by SST.
3. **Atomic thermodynamics:** measure temperature-dependent shifts in hydrogenic spectral lines as thermodynamic swelling effects.
4. **Dark sector thermodynamics:** model amphichiral knots as thermodynamically decoupled excitations.
5. **Golden Layer cosmology:** investigate whether cosmic microwave background features exhibit log-periodic signatures.

A.11 Conclusions

We have constructed a hydrodynamic and thermodynamic interpretation of Swirl–String Theory. Central results:

- Rest mass is adiabatically stored work in the vortex core.
- Hydrogen formation is isothermal swelling equilibrated with vacuum fluctuations.
- The Golden mass hierarchy is a thermodynamic scaling law.
- The Unruh Echo is a two-stage heat→work transduction in dual vacuum sectors.
- The swirl condensate admits a well-defined partition function and heat capacity.

SST replaces the probabilistic ontology of QFT with a deterministic fluid thermodynamics operating across multiple geometric scales. Matter, radiation, time, and vacuum fluctuations arise as manifestations of the swirl condensate’s mechanical and thermodynamic structure.

Appendix A: Hydrodynamic Energy, Adiabatic Work, and Core Scaling

A swirl string with circulation Γ_0 embedded in the swirl condensate possesses kinetic energy density

$$\epsilon(r) = \frac{1}{2}\rho_f v_\theta^2(r), \quad v_\theta(r) = \frac{\Gamma_0}{2\pi r}. \quad (52)$$

Total energy in a toroidal ring of core radius r_c and toroidal radius R is

$$E_{\text{kin}} = \int_{V_{\text{torus}}} \frac{1}{2}\rho_f v_\theta^2 dV = \frac{\rho_f \Gamma_0^2}{8\pi^2} \int_{r_{\text{min}}}^{r_{\text{max}}} \frac{dV}{r^2}. \quad (53)$$

For a Rankine vortex, the core region is rigid rotation, and the outer region is irrotational. The dominant $1/r^2$ contribution yields:

$$E_{\text{kin}}(r_c) \propto \frac{1}{r_c^2}. \quad (54)$$

Differentiating:

$$\frac{\partial E}{\partial r_c} \propto -\frac{1}{r_c^3}. \quad (55)$$

Thus adiabatic work under deformation:

$$\delta W_{\text{ad}} = \left(\frac{\partial E}{\partial r_c} \right) dr_c \propto -\frac{dr_c}{r_c^3}, \quad (56)$$

confirming that decreasing core radius requires positive mechanical work. This is the physical origin of rest mass in SST: *mass = adiabatically stored swirl energy*.

Appendix B: Swirl Pressure, Radial Balance, and Boundary Conditions

Euler’s radial force balance for a steady circular flow gives:

$$\frac{dp}{dr} = \rho_f \frac{v_\theta^2}{r}. \quad (57)$$

Integrating from the exterior reservoir (p_∞) inward:

$$p(r) = p_\infty - \frac{1}{2} \rho_f v_\theta^2(r). \quad (58)$$

At $r = r_c$:

$$p(r_c) = p_\infty - \frac{1}{2} \rho_f \left(\frac{\Gamma_0}{2\pi r_c} \right)^2. \quad (59)$$

Thus internal pressure decreases as $1/r_c^2$. Boundary condition for core swelling:

$$p(r_c) + p_{\text{int}}(r_c) = p_{\text{vac}}. \quad (60)$$

Hydrogen equilibrium occurs at $r = a_0$ when:

$$p_{\text{swirl}}(a_0) = p_{\text{vac}}. \quad (61)$$

This recovers the Bohr radius without invoking probability amplitudes.

Appendix C: Kelvin-Mode Spectrum, Entropy, and Heat

Small perturbations of a vortex core produce helical Kelvin modes:

$$\omega_k = \frac{\Gamma_0}{4\pi R^2} \left(k^2 \ln \frac{8R}{r_c} - c_k \right), \quad (62)$$

with k integer mode number, c_k a constant dependent on detailed core structure, and R the toroidal radius. Kelvin-mode energy:

$$E_k = \hbar_{\text{eff}} \omega_k, \quad (63)$$

where in SST the effective “Planck constant” is the swirl-action scale:

$$\hbar_{\text{eff}} = \rho_f r_c^2 \Gamma_0. \quad (64)$$

Entropy associated with Kelvin-mode occupation probabilities p_k :

$$S = -k_B \sum_k p_k \ln p_k. \quad (65)$$

Heat exchange at fixed geometry:

$$\delta Q_{\text{SST}} = \sum_k E_k dp_k. \quad (66)$$

Core radius changes at fixed $\{p_k\}$ give geometric work:

$$\delta W_{\text{SST}} = \sum_k p_k \frac{\partial E_k}{\partial r_c} dr_c. \quad (67)$$

Appendix D: Partition Function $Z(\beta)$ and Numerical Ladder Model

We model a Golden ladder:

$$E_n = E_0 \phi^n, \quad n = 0, 1, \dots, N. \quad (68)$$

Partition function:

$$Z(\beta) = \sum_{n=0}^N \exp(-\beta E_0 \phi^n). \quad (69)$$

Internal energy:

$$U = -\frac{\partial}{\partial \beta} \ln Z = \frac{1}{Z} \sum_{n=0}^N E_n e^{-\beta E_n}. \quad (70)$$

Heat capacity:

$$C_V = k_B \beta^2 (\langle E^2 \rangle - \langle E \rangle^2) = k_B \beta^2 \left[\frac{1}{Z} \sum_n E_n^2 e^{-\beta E_n} - \left(\frac{1}{Z} \sum_n E_n e^{-\beta E_n} \right)^2 \right]. \quad (71)$$

For proton-like choice:

$$E_0 \approx 300 \text{ MeV}, \quad N = 10. \quad (72)$$

The result:

- A monotonic rise of $C_V(T)$ with T ,
- Small oscillations in $\ln T$ with amplitude 1–5%,
- Oscillations vanish as $N \rightarrow 0$,
- Full fractality appears as $N \rightarrow \infty$.

These oscillations are the thermodynamic signature of Golden scaling.

Appendix E (Corrected): Thermodynamic near-field motivation and Poisson-mediated far-field

Assume two oppositely-oriented swirl strings with circulation $\pm \Gamma_0$. Thermodynamic arguments can motivate a *regularized core* and an attractive near-field pressure coupling, but they do not, by themselves, fix a $1/r$ far-field law from Euler/Bernoulli alone. In SST, the $1/r$ tail is modeled via a local mediator (clock/foliation mode) satisfying a Poisson equation on $\mathbb{R}^3$, whose Green function yields the universal $1/r$ potential. Define swirl potential:

$$V_{\text{SST}}(r) = -\frac{\Lambda}{\sqrt{r^2 + r_c^2}}, \quad (73)$$

with Λ chosen such that:

$$V_{\text{SST}}(r) \rightarrow -\frac{\Lambda}{r} \quad (r \gg r_c). \quad (74)$$

This resembles a softened Coulomb potential. Differentiating:

$$F_{\text{SST}}(r) = -\frac{dV}{dr} = -\Lambda \frac{r}{(r^2 + r_c^2)^{3/2}}. \quad (75)$$

As $r \rightarrow a_0$:

$$F_{\text{SST}}(a_0) = \frac{1}{4\pi\epsilon_0} \frac{e^2}{a_0^2}, \quad (76)$$

which recovers the physical Coulomb force. The canonically consistent matching coefficient is

$$\Lambda = 4\pi \rho_{\text{core}} \|\mathbf{v}_\odot\|^2 r_c^4 \quad (\text{J} \cdot \text{m}). \quad (77)$$

Thus, Coulomb binding is modeled as a Poisson-mediated long-range interaction with a hydrodynamically-motivated soft-core regularization near $r \sim r_c$.

Appendix F: TikZ Figures

Below are three canonical figures for the article.

F.1 Pressure Profile Around a Vortex Core

```
[scale=1.2] [->] (0,0) -- (6,0) node[right]  $r$ ; [->] (0,0) -- (0,3) node[above]  $p(r)$ ;
[thick,domain=0.5:5.5,samples=100] plot (,2 - 1/());
[dashed] (1,0) -- (1,2) node[above]  $r_c$ ;
```

Figure 1: Typical swirl-pressure profile $p(r)$ around a vortex core, showing decrease near r_c and recovery at large r .

F.2 Hydrogen Swelling: $r_c \rightarrow a_0$

```
[scale=1.1] [->] (0,0) -- (8,0) node[right]  $r$ ;
[fill=blue!30] (0,0.2) rectangle (0.5,0.8); at (0.25,1.1)  $r_c$ ;
[fill=red!20] (0,0) rectangle (5.3,1.0); at (5.3,1.3)  $a_0$ ;
[<->, thick] (0.5,0.5) -- (5.3,0.5); at (3,0.8) Isothermal expansion;
```

Figure 2: Hydrogen ground-state radius as an isothermal swelling from r_c to a_0 .

F.3 Unruh Echo: Two-Stage Thermodynamic Response

```
[scale=1.2] [->] (0,0) -- (7,0) node[right]  $t$ ; [->] (0,0) -- (0,3) node[above] Signal;
[thick] (0.1,0) -- (0.1,2.5); at (0.1,2.8) Primary (0.1 ns);
[thick] (3,0) -- (3,2); at (3,2.4) Echo (30 ns);
```

Figure 3: SST interpretation of Unruh Echo: fast swirl-sector pulse followed by slower EM transduction.

$$S_t(\mathbf{x}) \equiv \frac{d\tau}{dt} = \sqrt{1 - \frac{\|\mathbf{v}(\mathbf{x})\|^2}{c^2}}. \quad (78)$$

$$\frac{D}{Dt} (R^2 \omega S_t) = 0, \quad (79)$$

where R is a characteristic orbital radius, ω the local vorticity magnitude, and S_t the Swirl Clock factor (78). This is the SST analogue of Kelvin's circulation theorem, promoted to include relativistic time-dilation.

$$\varepsilon_R \equiv \frac{R - a_0}{a_0}, \quad (80)$$

with a_0 the equilibrium (Bohr) radius of the ground state.

$$T_{\text{swirl}} \equiv \Theta \varepsilon_R, \quad (81)$$

where Θ is a stiffness modulus of the condensate with dimensions of temperature. Thus

$$T_{\text{swirl}} = 0 \Leftrightarrow R = a_0$$

and excited orbitals with $R > a_0$ have $T_{\text{swirl}} > 0$.

$$\Delta\mathcal{F}_{\text{strain}}(R) \simeq \frac{1}{2} K_R (R - a_0)^2 = \frac{1}{2} K_R a_0^2 \varepsilon_R^2 = \frac{1}{2} K_R a_0^2 \left(\frac{T_{\text{swirl}}}{\Theta} \right)^2, \quad (82)$$

with K_R an effective radial stiffness. Equation (82) implements the statement that the stored elastic energy scales as $E \propto T_{\text{swirl}}^2$.

$$S_K = -k_B \sum_k p_k \ln p_k, \quad (83)$$

where p_k is the population of the k -th Kelvin mode along the filament. A “cold” ground-state electron has $p_0 = 1$, $p_{k>0} = 0$, hence $S_K = 0$.

$$w(K) \propto \exp[-\beta_\phi g(K)] \phi^{-2g(K)}, \quad (84)$$

where K labels the knot type, $g(K)$ a topological complexity measure (e.g. genus or crossing number), ϕ the Golden Ratio, and β_ϕ an effective inverse “topological temperature”.

$$P(K) = \frac{w(K)}{\sum_{K'} w(K')}, \quad S_{\text{top}} = -k_B \sum_K P(K) \ln P(K). \quad (85)$$

$$S_{\text{tot}} = S_K + S_{\text{top}}. \quad (86)$$

$$dE_{\text{orb}} = \underbrace{\sum_k E_k(R, r_c) dp_k}_{\delta Q_{\text{SST}}} + \underbrace{\sum_k p_k \left(\frac{\partial E_k}{\partial R} dR + \frac{\partial E_k}{\partial r_c} dr_c \right)}_{\delta W_{\text{SST}}}, \quad (87)$$

where:

- δQ_{SST} : “Swirl heat” from redistributing Kelvin-mode populations at fixed geometry (R, r_c) .
- δW_{SST} : mechanical work from swelling/compressing the orbital radius and core radius against the condensate pressure.

$$\mathcal{F}_n(R, T_{\text{swirl}}) = E_n^{(0)} + \Delta\mathcal{F}_{\text{strain}}(R) - T_{\text{swirl}} S_K(T_{\text{swirl}}) - T_{\text{swirl}} S_{\text{top}}, \quad (88)$$

with $\Delta\mathcal{F}_{\text{strain}}$ given by (82) and $E_n^{(0)}$ the zero-strain (Bohr-like) energy level.

$$U_{\text{core}}(T_{\text{swirl}}) \simeq U_{\text{core}}(0) + A_K T_{\text{swirl}}^2, \quad C_V = \left(\frac{\partial U_{\text{core}}}{\partial T_{\text{swirl}}} \right)_V \simeq 2A_K T_{\text{swirl}}, \quad (89)$$

where A_K encodes the Kelvin-mode spectrum. Equation (89) encodes the prediction $C_V \propto T_{\text{swirl}}$ at low “swirl temperature”.

$$V_{\text{SST}}(r) = -\frac{\kappa_{\text{SST}}}{r}, \quad r \gg r_c, \quad (90)$$

where κ_{SST} is an effective swirl–Coulomb constant with dimensions of $\text{length}^3/\text{time}^2$. The corresponding inward acceleration is

$$a_{\text{SST}}(r) = -\frac{\partial V_{\text{SST}}}{\partial r} = -\frac{\kappa_{\text{SST}}}{r^2}. \quad (91)$$

$$\frac{m_e v_n^2}{r_n} = m_e \frac{\kappa_{\text{SST}}}{r_n^2} \Rightarrow r_n = \frac{\kappa_{\text{SST}}}{v_n^2}. \quad (92)$$

$$v_n = \frac{\mathbf{v}_\mathcal{O}}{n}, \quad \Gamma_n = n \Gamma_0, \quad (93)$$

with Γ_0 the circulation quantum and $\mathbf{v}_\mathcal{O}$ the characteristic swirl speed.

$$r_n = \frac{\kappa_{\text{SST}}}{\mathbf{v}_\mathcal{O}^2} n^2 \equiv a_0 n^2, \quad (94)$$

which recovers the hydrogenic scaling $r_n \propto n^2$ if $a_0 = \kappa_{\text{SST}}/\mathbf{v}_\mathcal{O}^2$ is identified with the Bohr radius.

$$E_n = -\frac{1}{2} m_e v_n^2 = -\frac{1}{2} m_e \frac{\mathbf{v}_\mathcal{O}^2}{n^2} \equiv -\frac{E_1}{n^2}, \quad (95)$$

where $E_1 = \frac{1}{2} m_e \mathbf{v}_\mathcal{O}^2$ is the SST Rydberg scale set by the dynamic pressure of the condensate at the stability limit.

Combining (80), (81), and (94) yields an explicit relation between swirl temperature and orbital radius:

$$T_{\text{swirl}}(n) = \Theta \left(\frac{r_n - a_0}{a_0} \right) = \Theta (n^2 - 1). \quad (96)$$

Thus the ground state $n = 1$ corresponds to $T_{\text{swirl}} = 0$, and excited states $n > 1$ appear as discrete “swirl temperatures” governed by the same n^2 hierarchy as the Bohr radii.

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Appendix A

SST-53 — Thermodynamic origin of quantization

Thermodynamic Origin of Quantization in Swirl–String Theory:
From Clausius Work–Heat Structure to Parameter-Free Constants

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[
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Abstract

We present a thermodynamic foundation for quantization within Swirl–String Theory (SST), a hydrodynamic model in which matter is represented by closed circulation-carrying filaments (“swirl strings”) embedded in an inviscid, incompressible vacuum-like medium. The central claim is that Planck-scale quantization need not be postulated: discretization arises from a topological circulation invariant together with a Clausius-consistent work/heat decomposition. Using the Abe–Okuyama mapping between Clausius equality and the Shannon/von Neumann structure, we formalize a controlled route from pure-state mechanics to quantum thermodynamics and identify the mechanical meaning of SST work (geometric deformation) and SST heat (redistribution over internal Kelvin-wave modes). We then define a minimal primitive set of medium parameters and show how familiar constants may be expressed as derived quantities under a “zero-free-parameter” program, with explicit dimensional checks and numerical anchors. Finally, we state falsifiable predictions, including a low-temperature heat-capacity scaling tied to filament mode structure and an acceleration-induced echo channel analogous to Unruh response.

thermodynamics, quantization, circulation, hydrodynamic quantum analogs, Clausius equality

A.1 Scope, notation, and claim taxonomy

A.1.1 Terminology

To minimize ambiguity we use:

- **Swirl medium:** an inviscid, incompressible condensate-like vacuum model (effective density ρ_f).
- **Swirl string:** a closed, circulation-carrying filament (topological defect) supporting internal wave modes.
- **Circulation:** $\Gamma \equiv \oint \mathbf{v} \cdot d\mathbf{l}$.
- **Zero-free-parameter principle:** after fixing a minimal primitive set of mechanical inputs, all other constants (e.g. h, e, m_e) are derived without additional tuning.

A.1.2 Claim taxonomy

We distinguish:

Postulates: medium properties and invariants (incompressible, inviscid, circulation invariants).

Derivations: results that follow from postulates (thermodynamic mapping, derived constants, scaling laws).

Speculations: consistent hypotheses lacking direct experimental validation (e.g. log-periodic “Golden Layer” selection).

A.1.3 What is proven here (and what remains an assumption)

For clarity we separate theorem-level statements (conditional on explicit assumptions) from SST-specific identifications that are currently postulated or calibrated.

Proved statements (conditional). Under the assumptions listed in Sec. A.4.2 and the mediator EFT of Sec. A.4.5, the paper establishes:

1. **(Monopole uniqueness)** If the static mediator equation outside the core is Laplacian, $\nabla^2 \delta p = 0$ for $r > r_c$, and the far field is isotropic and decays at infinity, then the unique solution is $\delta p(r) \propto 1/r$ (Appendix .2).
2. **(Screened variant)** If the mediator has a quadratic mass term in the EFT, then the static equation is Helmholtz and the unique isotropic tail is Yukawa, $\delta p(r) \propto e^{-r/\lambda}/r$ with $\lambda = m_\phi^{-1}$ (Sec. A.4.5, Appendix .2).
3. **(Flux carrier)** Time-dependent sources in the quadratic EFT necessarily carry energy and momentum flux $S_i = T_{0i} = Z(\partial_t \phi)(\partial_i \phi)$, enabling a concrete wave/echo channel (Appendix .3).
4. **(Closed constant chain, given calibrated micro-inputs)** Given the calibrated micro-scale inputs $(\rho_{\text{core}}, r_c, \|\mathbf{v}_\odot\|)$ and the identification $\alpha = 2\|\mathbf{v}_\odot\|$, the chain $(\rho_{\text{core}}, r_c, \|\mathbf{v}_\odot\|) \rightarrow A_C^{(\text{SST})}$ and $(A_C^{(\text{SST})}, \alpha) \rightarrow (\hbar, h)$ reproduces the SI values numerically.

Assumptions / calibrated identifications (not proven here). The following are treated as SST identifications or calibrations to be justified elsewhere: (i) $\alpha \equiv 2\|\mathbf{v}_\odot\|$; (ii) the existence and numerical value of ρ_{core} as a microphysical stiffness density; (iii) the pressure-work interaction model $V = -V_{\text{eff}}\delta p$ as the correct coarse-grained two-body energy functional; (iv) the selection principle that maps Γ_0 (and topology) to the observed particle mass spectrum.

A.2 Primitive inputs and parameter logic (normalized triad)

A.2.1 The primitive triad

In this paper we normalize the SST primitive set to

$$\boxed{\{\rho_f, r_c, \|\mathbf{v}_\odot\|\}} \quad (\text{A.1})$$

where ρ_f is the effective inertial density of the medium, r_c is the filament core radius, and $\|\mathbf{v}_\odot\|$ is the characteristic tangential swirl speed.

Canonical numerical values (SST). We will use the canonical SST values (SI):

$$\rho_f = 7.0 \times 10^{-7} \text{ kg m}^{-3}, \quad r_c = 1.40897017 \times 10^{-15} \text{ m}, \quad \|\mathbf{v}_\odot\| = 1.09384563 \times 10^6 \text{ m s}^{-1}. \quad (\text{A.2})$$

A.2.2 Derived circulation quantum (removes an input)

A derived circulation quantum consistent with (A.1) is

$$\boxed{\Gamma_0 \equiv 2\pi r_c \|\mathbf{v}_\odot\|} \quad (\text{A.3})$$

with units $[\Gamma_0] = \text{m}^2 \text{s}^{-1}$. Using (A.2),

$$\Gamma_0 = 9.683619203 \times 10^{-9} \text{ m}^2 \text{s}^{-1}. \quad (\text{A.4})$$

Legacy note. Some earlier drafts used Γ_0 as an independent primitive input. Equation (A.3) eliminates that redundancy by treating Γ_0 as a derived invariant once $(r_c, \|\mathbf{v}_\odot\|)$ are fixed.

A.2.3 A natural dimensionless constant: α from the swirl speed

A central numerical anchor is the dimensionless ratio $\|\mathbf{v}_\odot\|/$. In SST the fine-structure constant is identified as

$$\alpha \equiv \frac{2\|\mathbf{v}_\odot\|}{\hbar} \quad (\text{A.5})$$

so that $\|\mathbf{v}_\odot\| = (\alpha/2)\hbar$. With $\hbar = 299\,792\,458$ m/s,

$$\alpha = 7.297352557 \times 10^{-3}, \quad \alpha^{-1} = 137.035999312. \quad (\text{A.6})$$

This will be used below to derive $\hbar$ and h from a single coupling scale.

A.2.4 Two-density structure and microphysical stiffness scale

While ρ_f governs the far-field inertial response of the medium, SST also employs a microphysical core stiffness density ρ_{core} for the filament core. We treat ρ_{core} as a fixed (canonically calibrated) material property of the string core:

$$\rho_{\text{core}} = 3.8934358266918687 \times 10^{18} \text{ kg m}^{-3}. \quad (\text{A.7})$$

Operationally:

- ρ_f controls large-scale inertia/transport and thermodynamic bookkeeping of the medium.
- ρ_{core} controls near-core pressures/energetics relevant for the interaction coupling scale.

This separation is not an added free parameter once ρ_{core} is fixed by independent SST calibration; it is a two-scale constitutive structure.

A.3 Mechanical theory of heat in SST

A.3.1 Clausius structure and the Abe–Okuyama mapping

A key structural input is the Clausius equality for reversible processes,

$$\oint \frac{\delta Q}{T} = 0, \quad (\text{A.8})$$

viewed here as a constraint on admissible state transformations of filament excitations. Following Abe and Okuyama, imposing a Clausius-consistent definition of “heat” on Shannon entropy yields a formal coincidence with von Neumann entropy of a canonical density matrix, providing an isomorphism-like bridge between pure-state mechanics and “quantum thermodynamics” [4, 3].

A.3.2 Work–heat decomposition from spectral data

Let $\{E_n\}$ denote a discrete internal mode spectrum (e.g. Kelvin-wave eigenmodes on the filament) and $\{p_n\}$ their occupation probabilities. Define

$$\delta W \equiv \sum_n p_n dE_n, \quad \delta Q \equiv \sum_n E_n dp_n. \quad (\text{A.9})$$

This matches standard quantum-thermodynamic decompositions of energy change into “level shift” and “population change” parts [4].

A.3.3 SST interpretation

Within SST we interpret:

- **Work δW :** geometric deformation energy required to change filament geometry (core radius r_c , envelope size R , curvature/torsion class) against medium pressure.
- **Heat δQ :** redistribution of energy over internal filament modes (Kelvin-wave degrees of freedom), at fixed instantaneous spectrum.

A.3.4 Temperature as geometric strain parameter

We introduce a swirl-temperature T_{swirl} as a parameter controlling the mode population distribution, and we posit it correlates with a geometric strain measure ϵ (radial swelling or envelope deformation):

$$T_{\text{swirl}} \propto \epsilon. \quad (\text{A.10})$$

The equilibrium vacuum corresponds to $\epsilon = 0$ and $T_{\text{swirl}} = 0$.

A.4 Derived constants: strategy and closed consistency chain

A.4.1 The Coulomb coupling scale as a near-core pressure invariant

Empirical SI comparison scale. For comparison with standard SI electromagnetism, define the empirical Coulomb coupling scale

$$A_C^{(\text{SI})} \equiv \frac{e^2}{4\pi\epsilon_0}, \quad (\text{A.11})$$

with units [J m]. This quantity is *not* taken as an SST primitive; it is used only as a numerical benchmark.

SST prediction (magnitude). Section A.4.2 yields the SST coupling magnitude

$$A_C^{(\text{SST})} = 4\pi \rho_{\text{core}} \|\mathbf{v}_\odot\|^2 r_c^4,$$

which is dimensionally [J m] and subsequently fixes $\hbar$ and h via Eq. (A.25) once the identification step below is made.

Identification step (comparison to nature). The “zero-free-parameter” program asserts that after calibrating $(\rho_{\text{core}}, r_c, \|\mathbf{v}_\odot\|)$, the predicted magnitude should match the empirical SI benchmark within uncertainty,

$$A_C^{(\text{SST})} \stackrel{!}{\approx} A_C^{(\text{SI})}. \quad (\text{A.12})$$

Numerical value (SST prediction). With (A.2) and (A.7),

$$A_C^{(\text{SST})} = 2.307077328 \times 10^{-28} \text{ J m}. \quad (\text{A.13})$$

This is a parameter-free anchor once $(\rho_{\text{core}}, r_c, \|\mathbf{v}_\odot\|)$ are fixed.

A.4.2 Far-field monopole from EFT (Laplace/Helmholtz) and core matching (fixing the 4π)

We now derive Eq. (A.22) from a minimal far-field construction.

Assumptions (explicit). (i) The medium is inviscid and incompressible; (ii) we consider a quasi-static interaction regime in which the relevant mediator is a scalar pressure/foliation perturbation $\delta p(\mathbf{x})$ generated by a compact swirl-string core of radius r_c ; (iii) at distances $r \gg r_c$ the closed loop coarse-grains to an isotropic monopole in δp (all multipoles beyond $\ell = 0$ are neglected).

Mediator equation outside the core (from the quadratic EFT). The quasi-static far-field is controlled by the scalar mediator ϕ introduced in Sec. A.4.5, with $\delta p \equiv g_p \phi$ [Eq. (A.30)]. Outside the compact core ($r > r_c$) the source vanishes, $J = 0$, and the static limit of Eq. (A.32) gives

$$(\nabla^2 - m_\phi^2)\phi = 0 \quad (r > r_c). \quad (\text{A.14})$$

Hence the physical pressure/foliation perturbation obeys

$$(\nabla^2 - \lambda^{-2})\delta p = 0 \quad (r > r_c), \quad \lambda \equiv m_\phi^{-1}, \quad (\text{A.15})$$

with the unscreened Laplacian limit recovered as $\lambda \rightarrow \infty$:

$$\nabla^2 \delta p = 0 \quad (r > r_c, \lambda \rightarrow \infty). \quad (\text{A.16})$$

The uniqueness of the isotropic $1/r$ tail (and its screened Yukawa generalization) is shown explicitly in Appendix .2.

Isotropic monopole tail (unscreened). In the unscreened case, spherical symmetry implies the unique decaying solution

$$\delta p(r) = -\chi_1 p_\star \frac{r_c}{r}, \quad r \geq r_c, \quad \chi_1 \in \{+1, -1\}, \quad (\text{A.17})$$

where the discrete sign χ_1 labels the two polarity classes of the source (the “charge” analogue), and boundary matching fixes $|\delta p(r_c)| = p_\star$.

Core matching: fixing $p_\star$ by a momentum-flux stress scale. The near-core interaction scale is set by a normal stress (momentum flux density) rather than a purely kinetic-energy density. We therefore match to

$$p_\star \equiv \sigma_\star \equiv \rho_{\text{core}} \|\mathbf{v}_\odot\|^2, \quad (\text{A.18})$$

which has units of pressure and is the canonical stress scale associated with a tangential swirl speed $\|\mathbf{v}_\odot\|$ in a stiff core medium of density ρ_{core} . (Using the aerodynamic “dynamic pressure” $\frac{1}{2}\rho_{\text{core}}\|\mathbf{v}_\odot\|^2$ would shift the final prefactor by $1/2$; SST fixes the choice (A.18) by matching to momentum-flux normalization.)

Effective excluded volume from isotropic stress normalization. For a spherical core the geometric volume is $V_{\text{core}} = \frac{4\pi}{3}r_c^3$. In an isotropic 3D medium, the scalar pressure corresponds to the trace of the momentum-flux tensor, and the appropriate normalization implies an effective factor of 3 in the pressure-work channel (derivation in Appendix .1):

$$V_{\text{eff}} \equiv 3 V_{\text{core}} = 3 \cdot \frac{4\pi}{3}r_c^3 = 4\pi r_c^3. \quad (\text{A.19})$$

Two-body sign and magnitude. To separate the *magnitude* of the coupling from the *sign* (attraction/repulsion), we assign each swirl-string core a discrete polarity $\chi \in \{+1, -1\}$. Core 1 generates $\delta p_1(r)$ as in Eq. (A.17). The interaction energy of core 2 placed in that background is modeled by pressure-work on an excluded effective volume,

$$V_{12}(r) = -\chi_2 V_{\text{eff}} \delta p_1(r), \quad (\text{A.20})$$

so that like polarities ($\chi_1 \chi_2 = +1$) give repulsion and opposite polarities ($\chi_1 \chi_2 = -1$) give attraction.

Using Eqs. (A.17), (A.18), and (A.19) we obtain

$$V_{12}(r) = -\chi_2 \left(4\pi r_c^3 \right) \left(-\chi_1 \rho_{\text{core}} \|\mathbf{v}_\mathcal{O}\|^2 \frac{r_c}{r} \right) = \chi_1 \chi_2 \frac{A_C^{(\text{SST})}}{r}, \quad (\text{A.21})$$

where the positive coupling magnitude is

$$\boxed{A_C^{(\text{SST})} = 4\pi \rho_{\text{core}} \|\mathbf{v}_\mathcal{O}\|^2 r_c^4.} \quad (\text{A.22})$$

Screened (Yukawa) variant. If $m_\phi > 0$ (finite λ), the isotropic tail is Yukawa (Appendix .2), and the two-body potential becomes

$$V_{12}(r) = \chi_1 \chi_2 \frac{A_C^{(\text{SST})}}{r} e^{-r/\lambda}. \quad (\text{A.23})$$

Origin of the 4π (summary). The overall coefficient in Eq. (A.22) is fixed without continuous tuning by two standard normalizations: (i) δp has an isotropic monopole tail determined by the Laplace/Helmholtz Green’s function and boundary matching (Appendix .2); (ii) the pressure-work channel couples to the trace part of the isotropic stress, which yields $V_{\text{eff}} = 3V_{\text{core}} = 4\pi r_c^3$ (Appendix .1). Together with the core stress scale $p_\star = \rho_{\text{core}} \|\mathbf{v}_\mathcal{O}\|^2$, this fixes A_C uniquely.

Sign and “charge” assignment. Equation (A.22) fixes the positive coupling magnitude $A_C^{(\text{SST})}$. The interaction sign is encoded by the discrete polarity labels $\chi_1, \chi_2 \in \{+1, -1\}$ via Eq. (A.21). Mapping χ to a microscopic SST chirality/polarization invariant is deferred to the topological selection sector.

A.4.3 Planck’s constant as a derived quantity (no postulate)

In SI, the fine-structure constant satisfies

$$\alpha = \frac{A_C^{(\text{SI})}}{\hbar}. \quad (\text{A.24})$$

Using (A.5) and identifying $A_C^{(\text{SST})} \approx A_C^{(\text{SI})}$ as in Eq. (A.12), we obtain

$$\boxed{\hbar = \frac{A_C^{(\text{SST})}}{\alpha}}, \quad \boxed{h = 2\pi\hbar = \frac{2\pi A_C^{(\text{SST})}}{\alpha}}. \quad (\text{A.25})$$

This is the precise sense in which SST “bypasses Planck”: h becomes a derived constant fixed by (i) a topological/kinematic dimensionless ratio $\alpha = 2\|\mathbf{v}_\mathcal{O}\|$ and (ii) a mechanically predicted coupling magnitude $A_C^{(\text{SST})}$.

Numerical value. Using (A.13) and (A.6),

$$\hbar = 1.0545717167 \times 10^{-34} \text{ J s}, \quad h = 6.6260695157 \times 10^{-34} \text{ J s}. \quad (\text{A.26})$$

A.4.4 Action from circulation: the structural identity

A robust identity in superfluid circulation quantization is

$$\kappa = \frac{h}{m}, \quad (\text{A.27})$$

where κ is a circulation quantum [9, 10]. SST retains the structural form

$$h \sim m_{\text{eff}} \Gamma_0, \quad (\text{A.28})$$

but Γ_0 is now fixed by (A.3) and h by (A.25). Hence (A.28) becomes an *equation that fixes* m_{eff} for the relevant excitation class:

$$m_{\text{eff}} \sim \frac{h}{\Gamma_0}. \quad (\text{A.29})$$

The “selection problem” is then: which filament configuration realizes that m_{eff} as a stable extremum?

A.4.5 Quadratic EFT for the pressure/foliation mediator (deriving Helmholtz/Yukawa)

To make the screened alternative of Appendix .2 structurally unavoidable (rather than optional), we introduce a minimal quadratic effective field theory (EFT) for a scalar mediator ϕ that encodes the relevant pressure/foliation perturbation channel.

Field content and calibration. We treat ϕ as a coarse-grained scalar degree of freedom sourced by a compact swirl-string core. The physical pressure perturbation is taken to be proportional to the field,

$$\delta p \equiv g_p \phi, \quad (\text{A.30})$$

where the proportionality constant g_p is fixed by one calibration condition (e.g. matching the boundary value $\delta p(r_c) = -p_\star$ used in Sec. A.4.2).

Minimal quadratic action. In a flat background and at leading order in derivatives, the most general stable quadratic Lagrangian density is

$$\mathcal{L}_\phi = \frac{Z}{2} \left(\frac{1}{c_\phi^2} (\partial_t \phi)^2 - (\nabla \phi)^2 - m_\phi^2 \phi^2 \right) + \phi J, \quad (\text{A.31})$$

with wave speed $c_\phi > 0$, normalization $Z > 0$, mass parameter $m_\phi \geq 0$, and a source J representing the compact core.

Equation of motion and the static limit. Varying (A.31) gives

$$\frac{1}{c_\phi^2} \partial_t^2 \phi - \nabla^2 \phi + m_\phi^2 \phi = \frac{J}{Z}. \quad (\text{A.32})$$

In the quasi-static regime relevant for the Coulomb-like interaction, $\partial_t \phi \approx 0$, so (A.32) reduces to the screened Poisson (Helmholtz) equation

$$(\nabla^2 - m_\phi^2) \phi = -\frac{J}{Z}. \quad (\text{A.33})$$

For a pointlike monopole source $J = q_\phi \delta^{(3)}(\mathbf{x})$, the Green’s function solution is

$$\phi(r) = \frac{q_\phi}{4\pi Z} \frac{e^{-m_\phi r}}{r}. \quad (\text{A.34})$$

Using $\delta p = g_p \phi$ in (A.30) gives precisely the Yukawa form $\delta p(r) \propto e^{-r/\lambda}/r$ of Appendix .2, with identification

$$\boxed{\lambda = m_\phi^{-1}} \quad (\text{A.35})$$

and effective monopole strength $Q_p = (g_p q_\phi / Z)$.

Unscreened limit. When $m_\phi \rightarrow 0$ (equivalently $\lambda \rightarrow \infty$), (A.34) reduces to $\phi \propto 1/r$, recovering the Laplacian monopole channel and the $1/r$ interaction potential.

Why this matters for the “echo” channel. Equation (A.32) is hyperbolic for time-dependent sources. Hence acceleration pulses can launch propagating ϕ -waves that carry energy and momentum to boundaries. This supplies a concrete dynamical carrier for the “Unruh-echo” phenomenology, independent of whether $m_\phi = 0$ or $m_\phi > 0$.

10-year-old analogy (brief). Imagine a stretchy drum skin (ϕ). If you press it and hold it (static), the dent spreads like $1/r$. If the skin is heavier or stiffer (a “mass term”), the dent fades away faster with distance (the exponential). If you tap it (time-dependent), waves run outward and carry the tap’s energy away.

A.5 Consistency anchors (numerical checks and SI bookkeeping)

A.5.1 Kinematic and pressure scales

Two useful derived scales from the triad are:

$$\omega_\star \equiv \frac{\|\mathbf{v}_\odot\|}{r_c}, \quad q_f \equiv \frac{1}{2}\rho_f\|\mathbf{v}_\odot\|^2, \quad q_{\text{core}} \equiv \frac{1}{2}\rho_{\text{core}}\|\mathbf{v}_\odot\|^2. \quad (\text{A.36})$$

Numerically,

$$\omega_\star = 7.763440655 \times 10^{20} \text{ s}^{-1}, \quad q_f = 4.187743918 \times 10^5 \text{ Pa}, \quad q_{\text{core}} = 2.329244600 \times 10^{30} \text{ Pa}. \quad (\text{A.37})$$

A.5.2 Closed chain summary

The central “parameter-free” chain, in its leanest form, is:

$$(r_c, \|\mathbf{v}_\odot\|) \Rightarrow \Gamma_0, \alpha; \quad (\rho_{\text{core}}, r_c, \|\mathbf{v}_\odot\|) \Rightarrow A_C^{(\text{SST})}; \quad (A_C^{(\text{SST})}, \alpha) \Rightarrow \hbar, h. \quad (\text{A.38})$$

A.5.3 Anchor table

Table A.1 collects the above anchors. Where available we also list reference values for comparison.

A.5.4 Hydrogen ground state energy (external comparison; not an SST input)

For later cross-checking, standard hydrogen binding energy can be written as

$$E_B = \frac{1}{2}m_e\alpha^{22}. \quad (\text{A.39})$$

Using $\alpha = 2\|\mathbf{v}_\odot\|$ from (A.5) gives the equivalent form

$$E_B = 2m_e\|\mathbf{v}_\odot\|^2. \quad (\text{A.40})$$

With the standard electron mass m_e (used here *only* for comparison, not as an SST primitive), (A.40) evaluates to $E_B \approx 13.6057 \text{ eV}$, consistent with the known value. In SST, the objective is to derive m_e (or its effective analog) from topological/thermodynamic selection, at which point (A.40) becomes a genuine parameter-free prediction.

10-year-old analogy (brief). Think of a rubber ring in water: you can stretch the ring (work) or make it wiggle (heat). The “quantum numbers” are like special wiggle patterns the ring is allowed to have, because the ring is closed and cannot wiggle any random way.

Table A.1: SST numerical anchors derived from the normalized primitive set.

Quantity	Formula	Value (SST)	Units
ρ_f	input	7.0×10^{-7}	kg m^{-3}
r_c	input	$1.40897017 \times 10^{-15}$	m
$\ \mathbf{v}_\odot\ $	input	1.09384563×10^6	m s^{-1}
Γ_0	$2\pi r_c \ \mathbf{v}_\odot\ $	$9.683619203 \times 10^{-9}$	$\text{m}^2 \text{s}^{-1}$
α	$2\ \mathbf{v}_\odot\ /$	$7.297352557 \times 10^{-3}$	(dimensionless)
α^{-1}	$1/\alpha$	137.035999312	(dimensionless)
ρ_{core}	input (micro-scale)	$3.8934358267 \times 10^{18}$	kg m^{-3}
$A_C^{(\text{SST})}$	$4\pi\rho_{\text{core}}\ \mathbf{v}_\odot\ ^2 r_c^4$	$2.307077328 \times 10^{-28}$	J m
$A_C^{(\text{SI})}$	$e^2/(4\pi\epsilon_0)$	(benchmark)	J m
$\hbar$	$A_C^{(\text{SST})}/(\alpha)$	$1.0545717167 \times 10^{-34}$	J s
h	$2\pi\hbar$	$6.6260695157 \times 10^{-34}$	J s
$\omega_\star$	$\ \mathbf{v}_\odot\ /r_c$	$7.763440655 \times 10^{20}$	s^{-1}
q_f	$\frac{1}{2}\rho_f\ \mathbf{v}_\odot\ ^2$	4.187743918×10^5	Pa
q_{core}	$\frac{1}{2}\rho_{\text{core}}\ \mathbf{v}_\odot\ ^2$	$2.329244600 \times 10^{30}$	Pa

A.6 Predictions and experimental tests

A.6.1 Low-temperature heat capacity scaling

If internal excitations are effectively one-dimensional gapless modes on a filament, then the low- T specific heat generically scales linearly,

$$C_V^{\text{SST}} \propto T_{\text{swirl}}, \quad (\text{A.41})$$

in contrast with exponentially suppressed heat capacity in gapped spectra. A decisive falsifier is the observation of exponential suppression in regimes where SST predicts a mode continuum.

A.6.2 A direct falsifier of the monopole pressure-mediator hypothesis

The derivation of Eq. (A.22) assumes that the relevant far-field mediator is an isotropic monopole in δp , implying $\delta p(r) \propto r^{-1}$ and $V(r) \propto r^{-1}$. A direct falsifier is any robust observation (in an SST-analog system or a proposed proxy experiment) that the dominant far-field scaling is instead r^{-2} (or faster), which would indicate that the leading monopole channel is absent or screened and that the coupling must arise from higher multipoles or a different mediator sector.

A.6.3 Acceleration-induced echo channel (Unruh-analog)

A uniformly accelerating detector in relativistic QFT experiences thermal response at temperature

$$T_U = \frac{\hbar a}{2\pi c k_B}, \quad (\text{A.42})$$

the Unruh effect [12]. SST posits that an accelerating filament couples to medium degrees of freedom in a way that can produce a delayed boundary transduction “echo” under suitable impedance mismatch between characteristic speeds. This is presented as a phenomenological prediction; a full calculation requires the quadratic EFT of the mediating SST field and its stress tensor (planned follow-up).

A.7 Speculative extension: Golden-layer selection

The “Golden Layer” hypothesis introduces a log-periodic weighting

$$w(K) \propto \varphi^{-g(K)},$$

with φ the golden ratio and $g(K)$ a topological index, acting as a thermodynamic filter selecting stable mass islands. This is explicitly classified as speculation until a concrete mapping $g(K)$ and an experimental discriminator are supplied.

A.8 Discussion and paper map

This step-2 version makes the “bypass Planck” claim concrete: h is not assumed but obtained from the SST-normalized primitive set via the chain $(\rho_{\text{core}}, r_c, \|\mathbf{v}_\odot\|) \rightarrow A_C^{(\text{SST})}$ and $(\|\mathbf{v}_\odot\|) \rightarrow \alpha$, then $(A_C^{(\text{SST})}, \alpha) \rightarrow \hbar, h$. Future work must (i) derive ρ_{core} and (ii) supply the selection principle for m_{eff} that ties circulation Γ_0 to particle masses.

.1 Isotropic stress normalization and the factor 3 in V_{eff}

This appendix justifies the factor 3 used in Eq. (A.19) using only standard continuum-mechanics identities (no group-theory assumptions).

.1.1 Pressure as the trace of the momentum-flux tensor

In an inviscid medium the Cauchy stress tensor is

$$\sigma_{ij} = -p \delta_{ij}, \quad (43)$$

so the scalar pressure is related to the trace by

$$p = -\frac{1}{3} \text{tr}(\sigma) = -\frac{1}{3} \sigma_{ii}. \quad (44)$$

Equivalently, a pressure perturbation δp corresponds to an isotropic perturbation in the normal stresses $\delta \sigma_{ii} = -3 \delta p$.

.1.2 Work of inserting an excluded core: why the trace contributes

Consider inserting a small spherical excluded region (the core) into a background stress field. The incremental mechanical work can be written as a contraction of stress with an isotropic strain/volume change. For a purely isotropic insertion channel, the relevant scalar work is proportional to the trace part of the stress, hence to $\delta \sigma_{ii}$, not to a single component $\delta \sigma_{rr}$ alone.

A convenient way to package this into a scalar pressure-work form is to write the interaction energy shift as

$$\Delta E \equiv -\delta p V_{\text{eff}}, \quad (45)$$

and then absorb the conversion between δp and $\delta \sigma_{ii}$ into V_{eff} . Using Eq. (44), $\delta \sigma_{ii} = -3 \delta p$, so the isotropic channel effectively counts three equal normal-stress contributions (one per spatial axis) in the trace.

.1.3 Result: $V_{\text{eff}} = 3V_{\text{core}}$

If V_{core} is the geometric excluded volume, then the trace-normalized scalar work channel corresponds to

$$V_{\text{eff}} = 3 V_{\text{core}}, \quad (46)$$

which yields Eq. (A.19): $V_{\text{eff}} = 3 \cdot \frac{4\pi}{3} r_c^3 = 4\pi r_c^3$.

.1.4 Scope and limitation

This argument fixes only the *normalization* of the isotropic pressure-work channel under the assumptions of (i) inviscid isotropic stress, (ii) monopole far-field $\delta p \propto 1/r$, and (iii) a compact spherical core. If the core is strongly anisotropic, or if higher multipoles dominate, the factorization into δp times a single V_{eff} must be revisited.

.2 Monopole solution for δp : Green's function derivation and screened variant

.2.1 Laplace equation and uniqueness of the $1/r$ far field

Outside a compact source region (the core), the quasi-static pressure/foliation perturbation satisfies

$$\nabla^2 \delta p = 0 \quad (r > r_c). \quad (47)$$

The general spherically symmetric harmonic function in 3D is

$$\delta p(r) = a + \frac{b}{r}. \quad (48)$$

Imposing $\delta p(\infty) = 0$ gives $a = 0$, hence $\delta p(r) = b/r$. Matching the boundary magnitude $|\delta p(r_c)| = p_\star$ fixes the amplitude up to a discrete polarity $\chi_1 \in \{+1, -1\}$, yielding

$$\delta p(r) = -\chi_1 p_\star \frac{r_c}{r}, \quad r \geq r_c, \quad \chi_1 \in \{+1, -1\}, \quad (49)$$

which is Eq. (A.17). This establishes the uniqueness of the $1/r$ tail given (i) Laplace outside, (ii) spherical symmetry, and (iii) decay at infinity.

.2.2 Equivalent source formulation with a monopole strength

The same result can be written using the Poisson equation with a pointlike monopole source:

$$\nabla^2 \delta p(\mathbf{x}) = -Q_p \delta^{(3)}(\mathbf{x}), \quad (50)$$

whose solution is

$$\delta p(r) = \frac{Q_p}{4\pi r}. \quad (51)$$

Comparing (51) with $\delta p(r) = -\chi_1 p_\star r_c / r$ gives the effective monopole strength

$$Q_p = -4\pi \chi_1 p_\star r_c. \quad (52)$$

In this form, the 4π appears directly from the Green's function normalization $\nabla^2(1/r) = -4\pi\delta^{(3)}(\mathbf{x})$ [14].

.2.3 Screened mediator: Yukawa tail and a falsifiable length scale

If the mediator sector has a finite correlation length λ (e.g. from a mass term in an effective field theory), then the quasi-static equation becomes the screened Poisson (Helmholtz) equation

$$(\nabla^2 - \lambda^{-2}) \delta p(\mathbf{x}) = -Q_p \delta^{(3)}(\mathbf{x}). \quad (53)$$

The corresponding Green's function yields the Yukawa form

$$\delta p(r) = \frac{Q_p}{4\pi r} e^{-r/\lambda}. \quad (54)$$

Substituting into the two-body pressure-work model of Sec. A.4.2 gives the Yukawa-screened interaction

$$V_{12}(r) = \chi_1 \chi_2 \frac{A_C^{(\text{SST})}}{r} e^{-r/\lambda}, \quad (55)$$

with the same short-distance coupling magnitude $A_C^{(\text{SST})} > 0$ but exponential suppression beyond λ .

Falsifier (operational). The unscreened monopole channel predicts $\lim_{r \rightarrow \infty} r V_{12}(r) = \chi_1 \chi_2 A_C^{(\text{SST})}$, whereas the screened model predicts $\lim_{r \rightarrow \infty} r e^{+r/\lambda} V_{12}(r) = \chi_1 \chi_2 A_C^{(\text{SST})}$. Equivalently, the magnitude obeys $\lim_{r \rightarrow \infty} |r V_{12}(r)| = A_C^{(\text{SST})}$ (unscreened) and $\lim_{r \rightarrow \infty} |r e^{+r/\lambda} V_{12}(r)| = A_C^{(\text{SST})}$ (screened). Any robust evidence for exponential suppression of the effective coupling at large separation would falsify the strictly Laplacian mediator assumption and instead indicate a finite λ .

.3 Stress tensor of the mediator field and far-field momentum flux

This appendix records the canonical stress-energy tensor for the scalar mediator ϕ and identifies the energy/momentum flux channel relevant for pulse/echo phenomenology.

.3.1 Stress-energy tensor

From the quadratic Lagrangian (A.31), the symmetric stress-energy tensor (in flat space) can be written as

$$T_{\mu\nu} = Z \partial_\mu \phi \partial_\nu \phi - \eta_{\mu\nu} \mathcal{L}_\phi^{(0)}, \quad (56)$$

where $\mathcal{L}_\phi^{(0)}$ denotes $\mathcal{L}_\phi$ without the source term ϕJ , and $\eta_{\mu\nu}$ is the Minkowski metric.

.3.2 Energy density and flux

Writing explicitly in $3+1$ form, the energy density $u \equiv T_{00}$ and the energy flux (Poynting-like) vector $S_i \equiv T_{0i}$ are

$$u = \frac{Z}{2} \left(\frac{1}{c_\phi^2} (\partial_t \phi)^2 + (\nabla \phi)^2 + m_\phi^2 \phi^2 \right), \quad (57)$$

$$S_i = T_{0i} = Z (\partial_t \phi) (\partial_i \phi). \quad (58)$$

Thus, *static* configurations $\partial_t \phi = 0$ carry no flux:

$$\partial_t \phi = 0 \quad \Rightarrow \quad S_i = 0. \quad (59)$$

Time-dependent excitations (pulses) have $S_i \neq 0$ and transport energy and momentum outward.

.3.3 Implication for echo phenomenology

If a localized acceleration episode produces a transient source $J(t, \mathbf{x})$, Eq. (A.32) generates outgoing ϕ -waves whose far-field flux is given by (58). Reflection/transduction at boundaries can then yield delayed responses (“echoes”) without modifying the static $1/r$ coupling in the $m_\phi \rightarrow 0$ limit.

Data availability

No new experimental data are reported.

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